

CHAPTER 4

FORCES

1.5 Forces

1.5.1 Balanced and unbalanced forces

- 1 Identify and use different types of force, including weight (gravitational force), friction, drag, air resistance, tension (elastic force), electrostatic force, magnetic force, thrust (driving force) and contact force
- 2 Identify forces acting on an object and draw free-body diagram(s) representing the forces
- 3 State Newton's first law as 'an object either remains at rest or continues to move in a straight line at constant speed unless acted on by a resultant force'
- 4 State that a force may change the velocity of an object by changing its direction of motion or its speed
- 5 Determine the resultant of two or more forces acting along the same straight line
- 6 Recall and use the equation
resultant force = mass $\times$ acceleration
 $F = ma$
- 7 State Newton's third law as 'when object A exerts a force on object B, then object B exerts an equal and opposite force on object A'
- 8 Know that Newton's third law describes pairs of forces of the same type acting on different objects

1.5.2 Friction

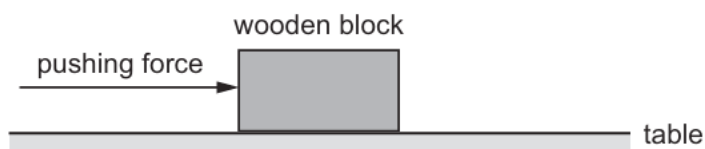
- 1 Describe friction as a force that may impede motion and produce heating
- 2 Understand the motion of objects acted on by a constant weight or driving force, with and without drag (including air resistance or resistance in a liquid)
- 3 Explain how an object reaches terminal velocity
- 4 Define the thinking distance, braking distance and stopping distance of a moving vehicle
- 5 Explain the factors that affect thinking and braking distance including speed, tiredness, alcohol, drugs, load, tyre surface and road conditions

1.5.4 Circular motion

- 1 Describe, qualitatively, motion in a circular path due to a force perpendicular to the motion as:
 - (a) speed increases if force increases, with mass and radius constant
 - (b) radius decreases if force increases, with mass and speed constant
 - (c) an increased mass requires an increased force to keep speed and radius constant
$$(F = \frac{mv^2}{r} \text{ is not required})$$

1. O/N/23/11/Q3

A wooden block is pushed across a table at constant speed.

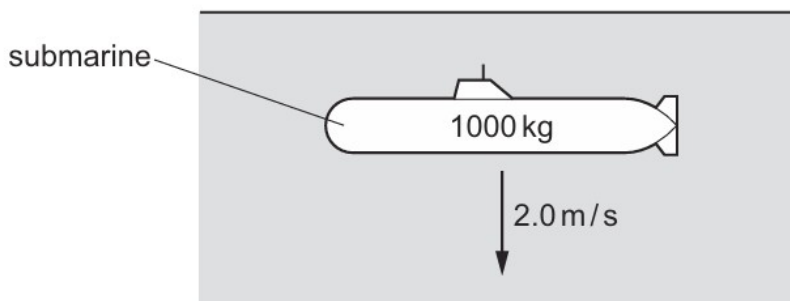


Which statement is correct?

- A** The frictional force increases as the block moves at constant speed.
- B** The frictional force is equal and opposite to the pushing force.
- C** The frictional force is greater than the pushing force.
- D** The frictional force is less than the pushing force.

2. O/N/23/12/Q5

A small submarine of mass 1000 kg sinks in water with a constant speed of 2.0 m/s.



What is the resultant force exerted on the submarine as it sinks?

- A** 0
- B** 500 N downwards
- C** 2000 N downwards
- D** 9800 N downwards

3. O/N/23/12/Q6

A student drops, from rest, a light ball in air. The ball experiences some air resistance as it falls.

What happens to the velocity and to the acceleration of the ball during the first few seconds after release?

	velocity	acceleration
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

4. O/N/23/12/Q7

The stopping distance of a car travelling at a certain speed is measured in four different situations.

- on a dry road with new tyres
- on a dry road with old, worn tyres
- on a wet road with new tyres
- on a wet road with old, worn tyres

The results are shown.

	W	X	Y	Z
stopping distance / m	12	16	17	40

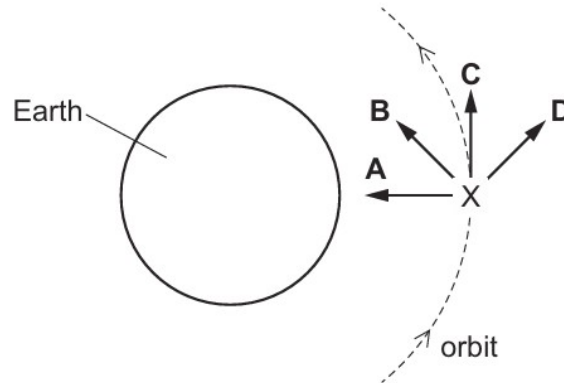
Which two results are possible measurements on the dry road with old, worn tyres?

- A** W and X **B** X and Y **C** X and Z **D** Y and Z

5. O/N/23/12/Q9

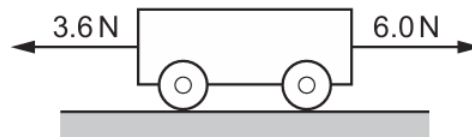
The International Space Station orbits the Earth.

Which arrow shows the direction of the force on the International Space Station when it is at point X?



6. M/J/23/11/Q7

The diagram shows the only two horizontal forces acting on a trolley that is accelerating at 0.12 m/s^2 .



What is the mass of the trolley?

- A** 0.050 kg **B** 0.13 kg **C** 20 kg **D** 80 kg

7. M/J/23/12/Q8

Newton's third law describes a pair of forces.

Which row shows whether the two forces are of the same type and whether they act on the same object?

	the types of forces in the pair	the objects on which they act
A	different	different
B	different	same
C	same	different
D	same	same

8. M/J/23/12/Q9

The stopping distance of a car is the sum of the thinking distance and the braking distance.

Which factors affect the braking distance?

	speed of car	tiredness of driver	condition of road
A	✓	✗	✗
B	✓	✓	✓
C	✓	✗	✓
D	✗	✓	✓

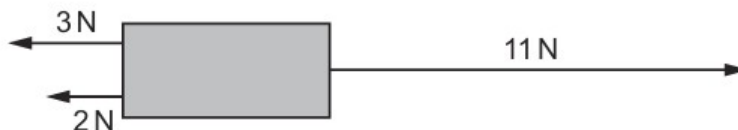
key

✓ = affects braking distance

✗ = does not affect braking distance

9. O/N/22/11/Q1

The diagram shows three forces acting on a block. The resultant force is 6 N to the right.



Which additional force produces a resultant force of 3 N to the left?

- A** 3 N to the left
- B** 9 N to the left
- C** 6 N to the right
- D** 13 N to the right

10. O/N/22/11/Q3

A car of mass 1000 kg is travelling down a steep hill. The brakes fail and the driver uses a horizontal sand-filled safety road to stop the car.

The car enters the sand at a speed of 10 m/s and experiences a constant stopping force of 2500 N.

How far does the car travel in the sand before coming to rest?

- A** 2.0 m **B** 4.0 m **C** 20 m **D** 40 m

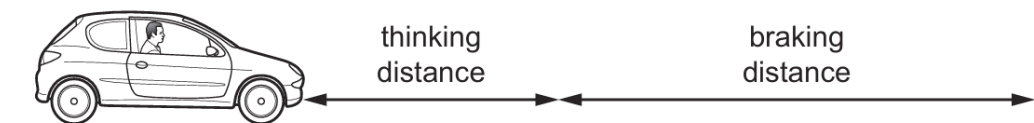
11. O/N/22/11/Q6

What causes the frictional effects experienced by a cyclist to increase?

- A** using a lighter bicycle
B travelling at greater speed
C a smoother tyre surface
D a road surface covered in oil

12. O/N/22/11/Q7

The diagram shows a car at the moment the driver sees a hazard ahead.



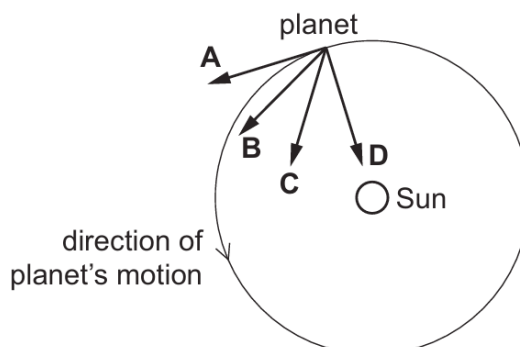
Which expression is equal to the stopping distance?

- A** braking distance + thinking distance
B braking distance – thinking distance
C braking distance $\times$ thinking distance
D braking distance $\div$ thinking distance

13. O/N/22/11/Q8

A planet orbits the Sun. The planet moves in a circular path at constant speed, but is accelerating because it is changing direction.

In which direction does the resultant force act to cause this acceleration?



14. O/N/22/12/Q6

A block of wood is placed on a table. External forces act on the block but the block remains stationary.

A student suggests three conditions for the block to remain stationary.

condition P The resultant force on the block **must** equal zero.

condition Q The resultant moment on the block **must** equal zero.

condition R The external forces **must** act through the centre of mass of the block.

Which conditions are necessary for the block to remain stationary?

A P and Q only **B** P and R only **C** Q and R only **D** P, Q and R

15. O/N/22/12/Q8

A car travels on a horizontal road around a bend at constant speed.

What is the direction of the resultant force on the car?

A backwards

B forwards

C towards the centre of the bend

D towards the road

16. O/N/22/12/Q7

A car travels at 60km/h on a straight road. The road is dry. The driver applies the brakes suddenly. The table shows the thinking distance, the braking distance and the stopping distance.

thinking distance /m	braking distance /m	stopping distance /m
27	22	49

The same car is driven by the same driver at 60km/h on the same road when it is wet. The driver applies the brakes suddenly again.

What is the effect of the wet road on the thinking distance, the braking distance and the stopping distance?

	thinking distance	braking distance	stopping distance
A	no change	increases	increases
B	no change	no change	no change
C	increases	increases	increases
D	increases	no change	increases

17. M/J/22/11/Q2

A driver drives a car at uniform velocity along a road.

The driver sees a hazard and applies the brakes.

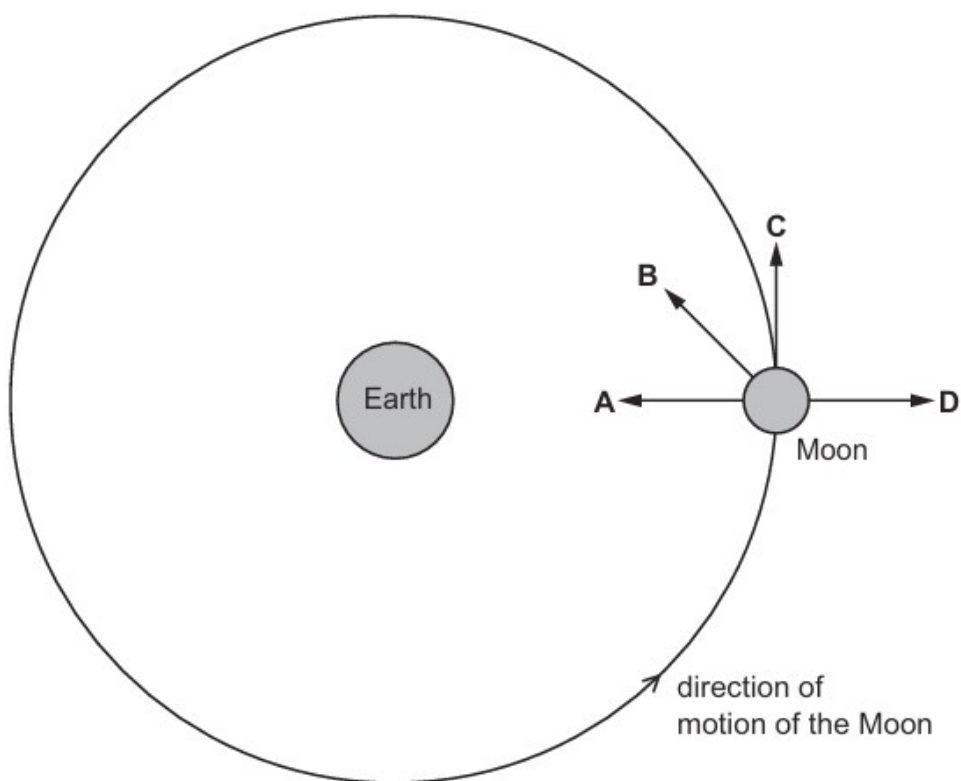
What is the car's motion as it moves through the thinking distance and as it moves through the braking distance?

	motion through the thinking distance	motion through the braking distance
A	deceleration	deceleration
B	deceleration	uniform velocity
C	uniform velocity	deceleration
D	uniform velocity	uniform velocity

18. M/J/22/11/Q3

The diagram represents the Moon in its orbit around the Earth.

Which arrow represents the direction of the resultant force acting on the Moon at the instant shown?



19. M/J/22/12/Q1

A force of 3.0 N and a force of 4.0 N act on an object.

What is the maximum possible resultant of these two forces?

- A** 1.0 N **B** 5.0 N **C** 7.0 N **D** 12 N

20. M/J/22/12/Q4

A 60 kg passenger enters a stationary lift. The gravitational field strength g is 10 N/kg.

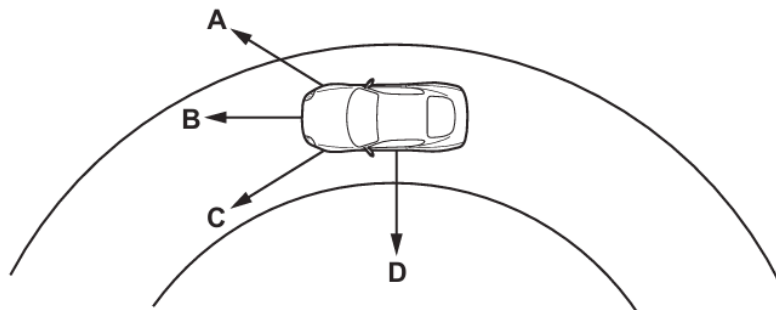
How much force does the floor of the lift exert on the passenger when the lift accelerates upwards at 2.0 m/s^2 ?

- A** 120 N **B** 480 N **C** 600 N **D** 720 N

21. O/N/21/11/Q4

The diagram shows a car going around a circular track at constant speed.

Which arrow shows the direction of the resultant force on the car?



22. O/N/21/11/Q5

An electron has mass.

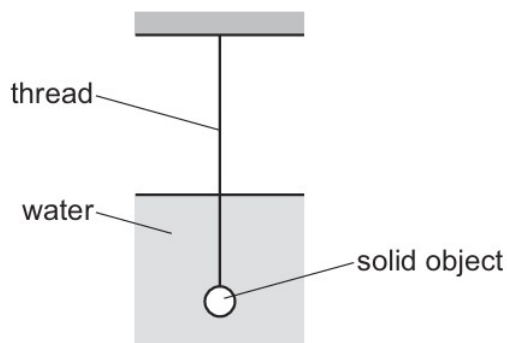
Where does a stationary electron experience a force?

- A** in an electric field only
B in a gravitational field only
C in a gravitational field and an electric field only
D in a gravitational, electric and magnetic field

23. O/N/21/12/Q4

A solid object, immersed in water, hangs from an elastic thread. Three forces act on the object: its weight W , the tension in the thread T , and a force F from the water.

The force F acts upwards.



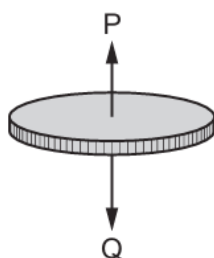
Which equation is correct when the object is stationary?

- A** $F + W = 0$
- B** $F - T = 0$
- C** $F - W - T = 0$
- D** $F - W + T = 0$

24. M/J/21/11/Q4

A coin falls from rest through the air and eventually reaches a constant speed.

There is a resultant force acting on the coin due to the two forces P and Q shown in the diagram.



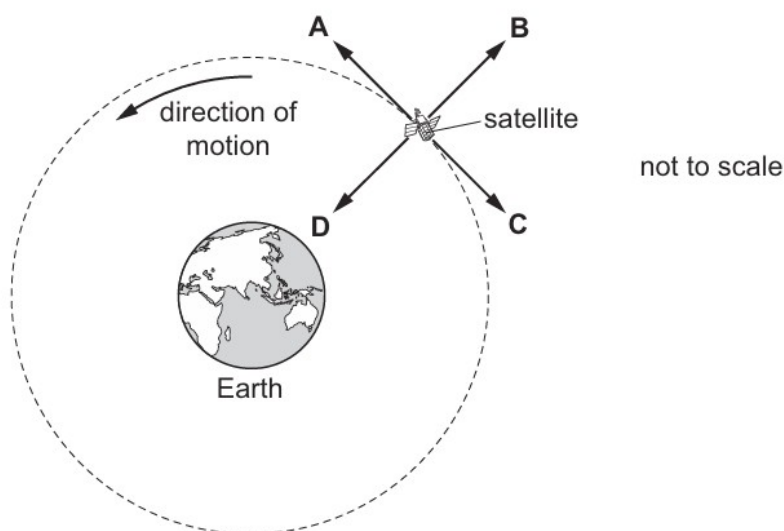
What happens to force P and what happens to the resultant force before the coin reaches constant speed?

	force P	resultant force
A	decreases	increases
B	decreases	decreases
C	increases	decreases
D	increases	increases

25. M/J/21/11/Q5

A satellite is shown moving around the Earth in a circular path at a constant speed.

Which arrow shows the direction of the force on the satellite?



26. M/J/21/12/Q4

A student lists three changes that affect the stopping distance of a car.

- 1 increasing the braking force
- 2 increasing the friction between the tyres and the road
- 3 increasing the speed of the car

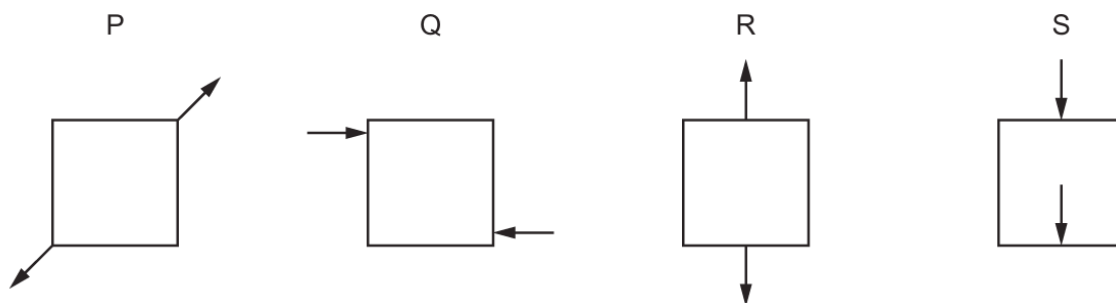
Which change or changes increase the stopping distance?

- A** 1 and 2 **B** 1 and 3 **C** 1 only **D** 3 only

27. M/J/21/12/Q3

The diagrams show the forces acting on four identical solid blocks.

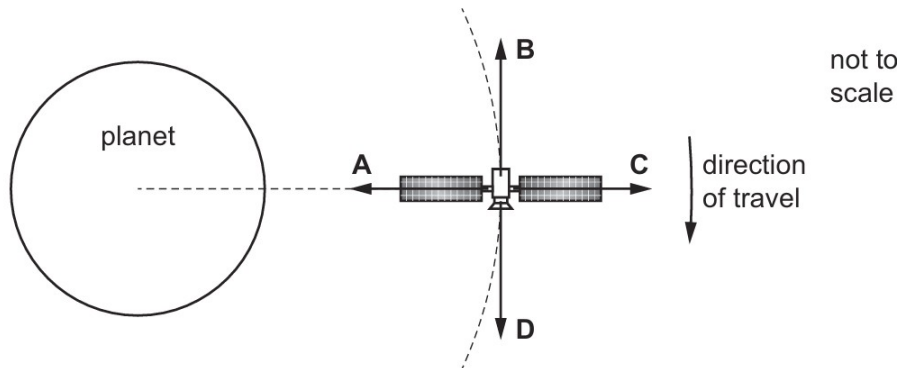
Each arrow represents a force of 20 N.



28. M/J/21/12/Q5

The diagram shows a satellite orbiting a planet at a steady speed.

In which direction does the resultant force act on the satellite?



29. O/N/20/11/Q3

A car of mass 800 kg has a forward acceleration of 2.5 m/s^2 . A frictional force of 1200 N opposes the motion of the car.

What is the driving force due to the engine of the car?

- A** 800 N **B** 1200 N **C** 2000 N **D** 3200 N

30. O/N/20/11/Q7

An electrically charged plastic ball is at rest.

Which types of field are caused by the ball?

- A** electric, gravitational and magnetic
B electric and gravitational only
C electric and magnetic only
D gravitational and magnetic only

31. O/N/20/11/Q4

A parachutist falling at a steady speed opens her parachute.

Which row is correct for the direction of the resultant force and for the direction of the acceleration of the parachutist just after her parachute opens?

	resultant force direction	acceleration direction
A	downwards	downwards
B	downwards	upwards
C	upwards	downwards
D	upwards	upwards

32. O/N/20/11/Q11

Which list contains only quantities that can be changed by a force?

- A mass, shape, velocity
- B mass, shape, volume
- C mass, velocity, volume
- D shape, velocity, volume

33. O/N/20/12/Q4

A car is travelling forwards at high speed. The brakes are applied and the car skids along the road surface.

In which direction and where does the friction force act?

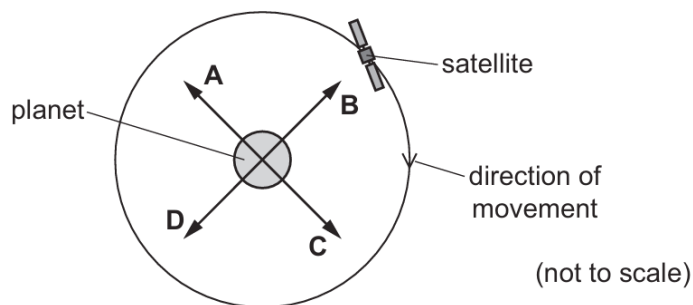
- A backwards on the tyres
- B forwards on the tyres
- C upwards on the tyres
- D downwards on the road

34. O/N/20/12/Q6

A satellite orbits a planet in a circular path as shown. It has constant speed.

There is a force on the planet due to the satellite.

In which direction is the force on the planet?



35. M/J/20/11/Q4

A car of weight 11 000 N moves with constant velocity along a horizontal road. A driving force of 5000 N acts on the car.

What is the force opposing the motion of the car?

- | | | | |
|----------|----------|------------|------------|
| A 5000 N | B 6000 N | C 11 000 N | D 16 000 N |
|----------|----------|------------|------------|

36. M/J/20/11/Q5

A man with an open parachute falls to Earth at constant speed. The following forces act:

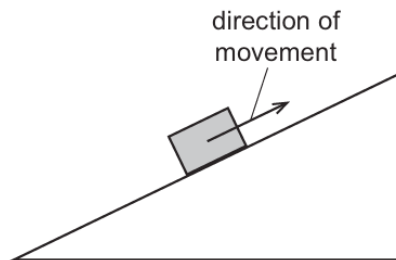
- P the upward force of the parachute on the man
- Q the upward force of the man on the Earth
- R the downward force of the Earth on the parachute
- S the downward force of the man on the parachute

Which two forces are a Newton's third law pair?

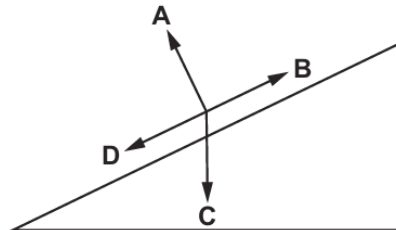
- A** P and Q **B** P and R **C** P and S **D** Q and R

37. M/J/20/11/Q6

A box is pulled up a rough slope, as shown.



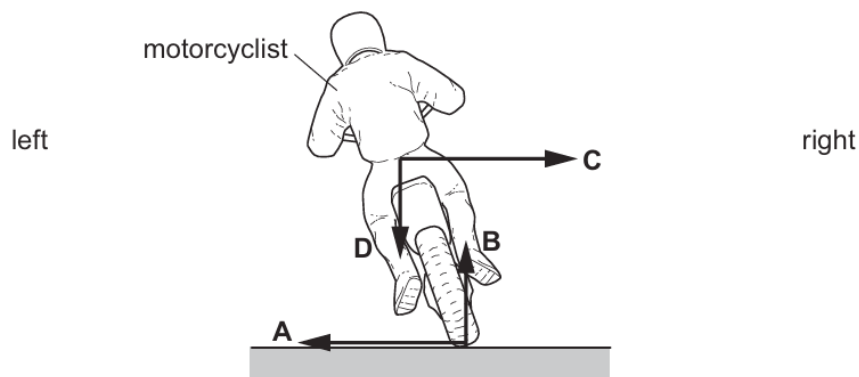
In which direction does friction act on the box?



38. M/J/20/11/Q7

The diagram shows a motorcyclist leaning over in order to turn the corner to the left.

Which force causes him to turn?



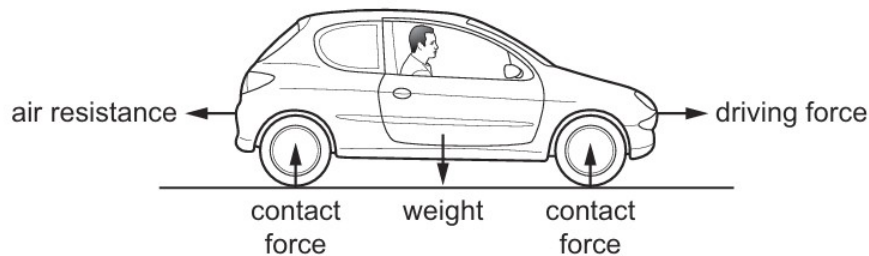
39. M/J/20/11/Q8

Which type of force causes the Earth to orbit the Sun?

- A elastic
- B electrostatic
- C gravitational
- D magnetic

40. O/N/19/11/Q6

A car is accelerating along a straight, horizontal road. The diagram shows forces acting on the car.



Which forces are balanced?

- A contact forces and air resistance
- B contact forces and weight
- C driving force and air resistance
- D driving force and weight

41. O/N/19/11/Q7

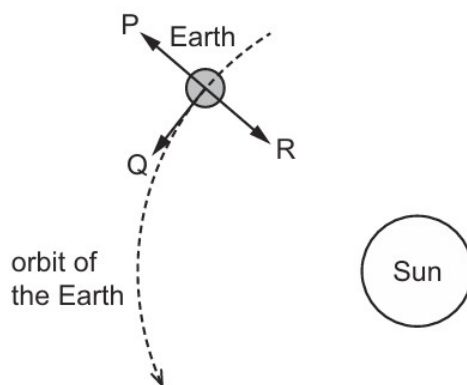
A single force is applied to a body.

What **cannot** happen?

- A a change in the direction of the moving body
- B a change in the mass of the body initially at rest
- C a change in the position of the body initially at rest
- D a change in the speed of the moving body

42. O/N/19/11/Q8

The Earth travels at constant speed in a circular orbit around the Sun.



Which arrows show the direction of the acceleration of the Earth and the direction of the velocity of the Earth?

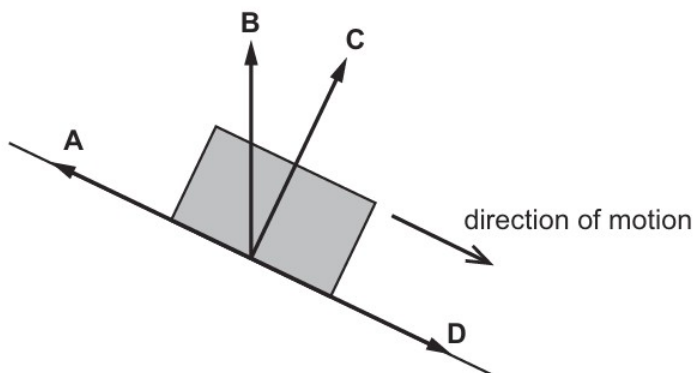
	direction of acceleration	direction of velocity
A	P	Q
B	Q	P
C	Q	R
D	R	Q

43. O/N/19/11/Q9

A body slides down a frictionless slope, as shown.

As the body presses on the surface, the surface pushes back on the body.

In which direction does the surface push back on the body?



44. M/J/19/11/Q5

Four of the gravitational forces that act between bodies in the Solar System are listed below.

- P the force on the Moon due to the Earth
- Q the force on the Earth due to the Sun
- R the force on the Earth due to the Moon
- S the force on the Moon due to the Sun

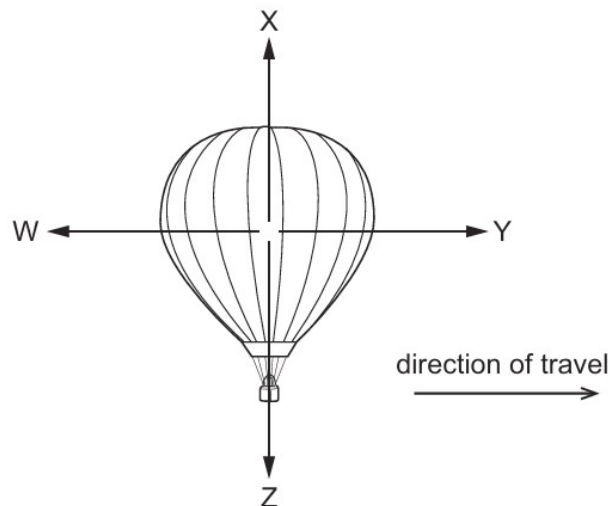
Which two forces are a Newton's third law pair (action and reaction)?

- A** P and Q **B** P and R **C** Q and S **D** R and S

45. M/J/19/11/Q6

A hot-air balloon is travelling at constant velocity and is at a constant height above the ground.

The diagram shows the only four forces acting on the balloon.



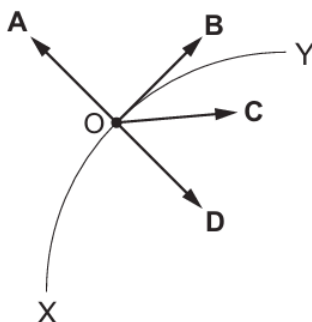
Which statement is correct?

- A** W and Y are equal, X and Z are equal.
- B** W and Y are equal, X is greater than Z.
- C** X and Z are equal, Y is greater than W.
- D** Y is greater than W. X is greater than Z.

46. M/J/19/11/Q7

The diagram represents an object O moving from X to Y along a circular path at constant speed.

What is the direction of the resultant force on O in the position shown?



47. M/J/19/12/Q6

A parachutist of mass 60 kg falls at a steady speed of 10 m/s. The gravitational field strength g is 10 N/kg.

What is the upward force acting on the parachutist?

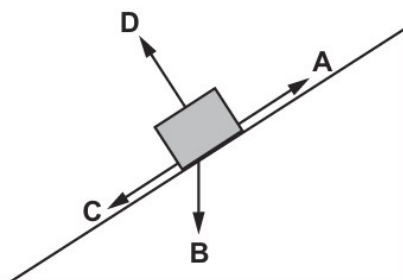
- A** 0 **B** 6.0 N **C** 60 N **D** 600 N

48. M/J/19/12/Q7

A force moves a box up a rough slope.

The diagram shows four forces acting on the box.

Which is the force due to friction on the box?



49. O/N/18/11/Q3

A skydiver is falling at terminal velocity.



Which row describes the acceleration of the skydiver and the velocity of the skydiver?

	acceleration of the skydiver	velocity of the skydiver
A	downwards	constant
B	downwards	zero
C	zero	constant
D	zero	zero

50. O/N/18/12/Q1

An apple falls from a tree.

Which row describes the forces acting on the apple and the motion of the apple as it falls?

	forces acting on the apple	motion of the apple
A	balanced	acceleration
B	balanced	constant speed
C	unbalanced	acceleration
D	unbalanced	constant speed

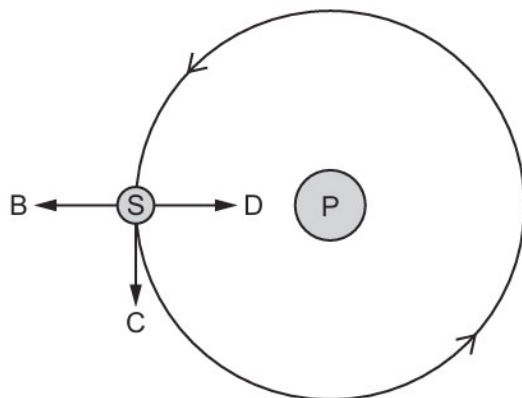
51. O/N/18/12/Q4

On which car is there a resultant force?

- A** a car moving along a straight horizontal road at constant speed
- B** a car moving around a bend at constant speed
- C** a car moving uphill at constant velocity
- D** a car that is stationary

52. M/J/18/11/Q3

The diagram shows a satellite S travelling at a constant speed in a circular orbit around a planet P.



Which statement is correct?

- A** The resultant force on the satellite is zero.
- B** The resultant force on the satellite is in direction B.
- C** The resultant force on the satellite is in direction C.
- D** The resultant force on the satellite is in direction D.

53. M/J/18/11/Q5

The minimum braking distance for a car is tested on a dry road.

The test is then repeated on a wet road.

What happens to the braking distance and to the frictional force between the tyres and the road?

	braking distance	frictional force
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

54. M/J/18/12/Q5

The planets in the Solar System orbit the Sun.

Which statement is correct?

- A** There is a force on each planet away from the Sun.
- B** There is a force on each planet in the direction in which it travels.
- C** There is a force on each planet opposite to the direction in which it travels.
- D** There is a force on each planet towards the Sun.

55. M/J/18/12/Q6

Which forces act on a skydiver who is falling at terminal velocity?

- A** air resistance and weight
- B** air resistance only
- C** weight only
- D** no forces act

56. M/J/18/12/Q11

A force of 1600 N accelerates a car, of mass 800 kg, from rest.

What is the car's acceleration and its velocity after 4.0 s?

	<u>acceleration</u> m/s ²	<u>velocity</u> m/s
A	0.50	0.13
B	0.50	0.50
C	2.0	2.0
D	2.0	8.0

57. O/N/17/11/Q5

A skydiver falls at terminal velocity. He then opens his parachute.

Which row gives the direction of the resultant force on the skydiver and the direction of the acceleration of the skydiver immediately after the parachute opens?

	resultant force	acceleration
A	downwards	downwards
B	downwards	upwards
C	upwards	downwards
D	upwards	upwards

58. O/N/17/12/Q4

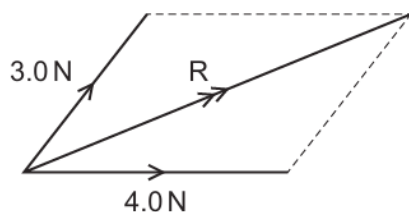
A free-fall skydiver jumps from a plane. As he falls there is a force acting upwards and a force acting downwards on his body. These produce a resultant force.

Before he reaches terminal velocity, how do the sizes of the forces change?

	downward force	upward force	resultant force
A	decreases	decreases	stays the same
B	increases	stays the same	decreases
C	stays the same	increases	decreases
D	stays the same	increases	increases

59. M/J/17/11/Q3

The diagram shows the resultant R of a 3.0 N force and a 4.0 N force acting at a point.



The angle between the 3.0 N force and the 4.0 N force can be any value from 0° to 90° .

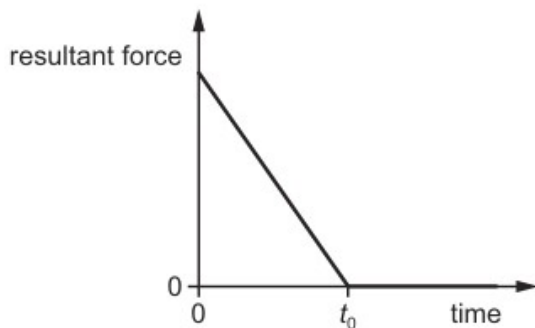
Which value of R is **not** possible?

- A** 4.0 N **B** 5.0 N **C** 6.0 N **D** 7.0 N

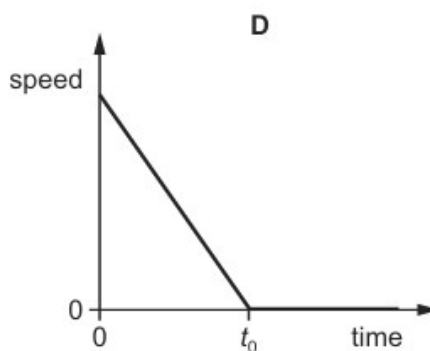
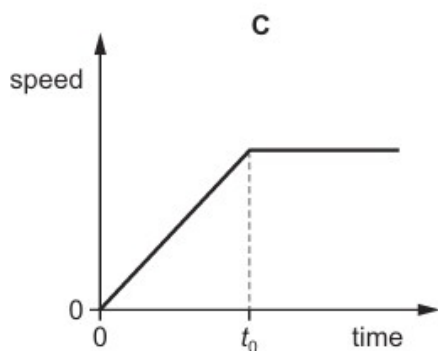
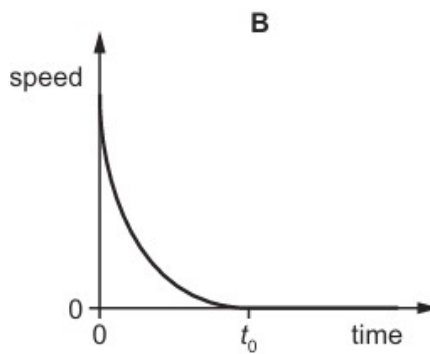
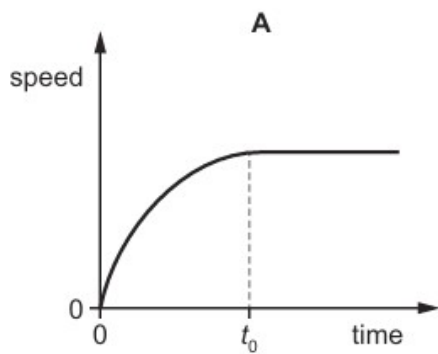
60. M/J/17/11/Q5

A resultant force acts on an object and causes it to move in a straight line.

The graph shows how the resultant force varies with time.



Which graph is the speed-time graph for the object?



61. M/J/17/11/Q6

A skydiver jumps from an aeroplane. After a few seconds, he reaches a terminal velocity without opening his parachute.

Why does he reach terminal velocity?

- A Air resistance becomes greater than his weight and slows him down.
- B Air resistance decreases and he speeds up.
- C Air resistance increases and balances his weight so that his acceleration is zero.
- D His weight decreases and balances the air resistance.

62. M/J/17/11/Q7

A satellite is orbiting the Earth in a circular orbit.

Which two quantities are always in the same direction as each other?

- A the acceleration of the satellite and the displacement of the satellite
- B the acceleration of the satellite and the velocity of the satellite
- C the resultant force on the satellite and the acceleration of the satellite
- D the resultant force on the satellite and the velocity of the satellite

63. M/J/17/11/Q8

A satellite is in a circular orbit around a planet.

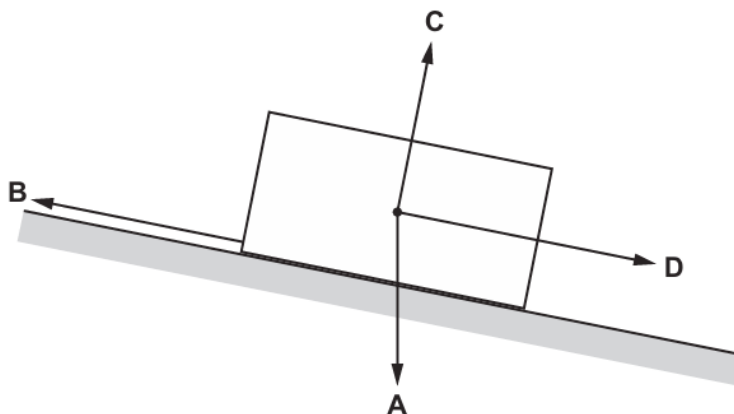
Which statement is correct?

- A Its acceleration is constant in direction but not in size.
- B Its acceleration is constant in size but not in direction.
- C Its gravitational potential energy varies.
- D Its velocity is constant.

64. M/J/17/11/Q9

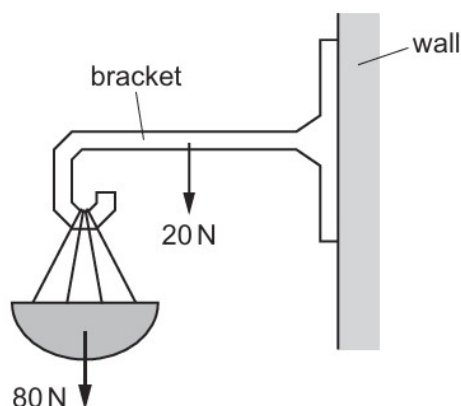
The diagram shows a block of wood resting on a sloping board.

Which arrow shows the direction of the gravitational force acting on the block?



65. M/J/17/12/Q6

A hanging basket is fixed to a wall by a bracket.



The weight of the basket is 80 N. The weight of the bracket is 20 N.

What is the size of the upwards force exerted on the bracket by the wall?

- A** 60 N **B** 70 N **C** 90 N **D** 100 N

66. M/J/17/12/Q7

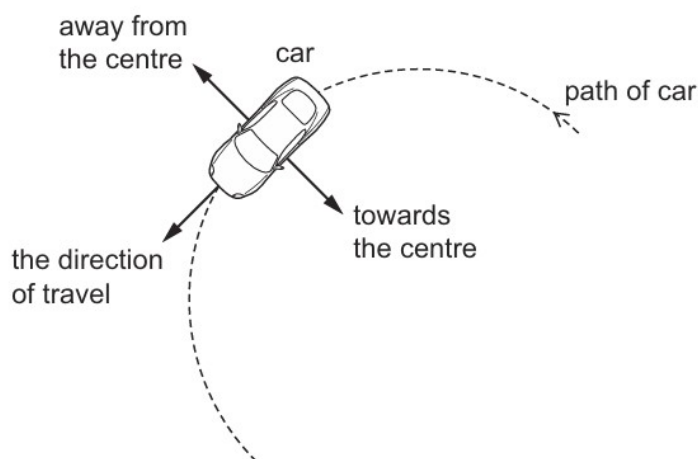
A train of mass 240 000 kg is travelling at a speed of 60 m/s. The brakes are applied and it decelerates for 10 minutes until it comes to rest.

What is the average resultant force?

- A** 24 000 N **B** 40 000 N **C** 480 000 N **D** 720 000 N

67. M/J/17/12/Q8

A car travels at a constant speed along a circular, horizontal path.



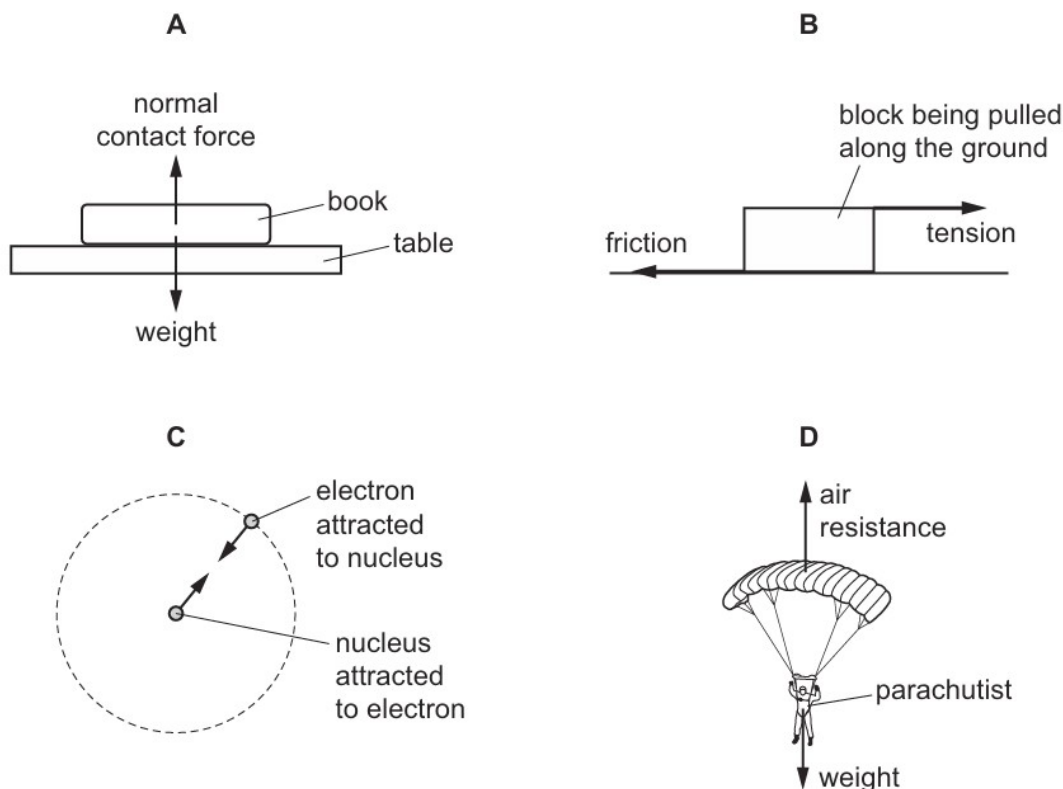
Which statement describes the forces acting on the car?

- A** They are balanced as the car is moving at constant speed.
B They are unbalanced with a resultant in the direction away from the centre.
C They are unbalanced with a resultant in the direction of travel of the car.
D They are unbalanced with a resultant in the direction towards the centre.

68. O/N/16/11/Q6

Each diagram shows two forces that are equal in magnitude.

Which diagram shows two forces that are a Newton's third law pair of forces?



69. O/N/16/12/Q3

Two aluminium spheres P and Q, of masses 0.50 kg and 1.0 kg respectively, are released at the same time from the same height.

Why do they have the same acceleration?

- A** The density of P is equal to the density of Q.
- B** The force on P is equal to the force on Q.
- C** The force on Q is double the force on P.
- D** The gravitational field strength for Q is double that for P.

70. O/N/16/12/Q5

A force of 20 N acts on an object of mass 5.0 kg in the forward direction. A second force also acts on the object. The forward acceleration of the object is 3.0 m/s^2 .

What is the size and direction of the second force?

	size of force / N	direction of force
A	5.0	backwards
B	15	backwards
C	15	forwards
D	35	forwards

71. M/J/16/11/Q5

A man pulls a sledge of mass 25 kg across level ground with a horizontal force of 60 N. A constant force of friction of 20 N acts on the sledge.

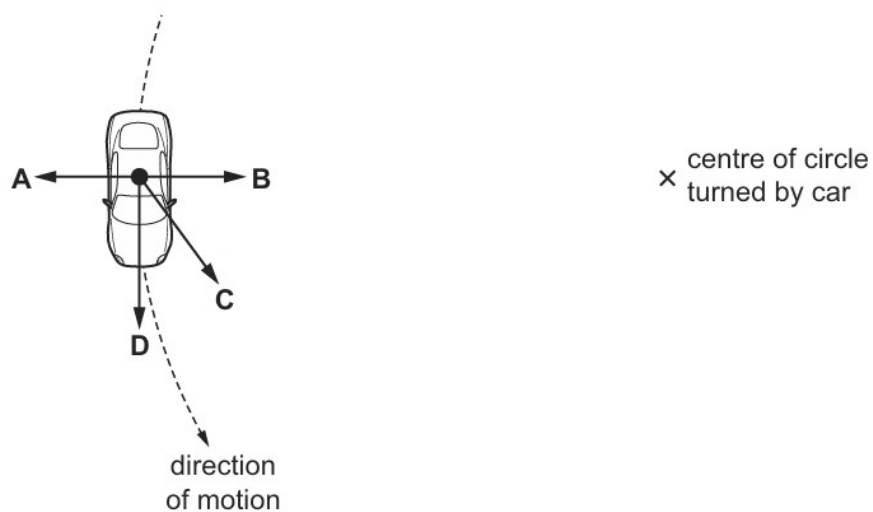
What is the acceleration of the sledge?

- A** 0.63 m/s^2 **B** 1.6 m/s^2 **C** 2.4 m/s^2 **D** 3.2 m/s^2

72. M/J/16/11//Q6

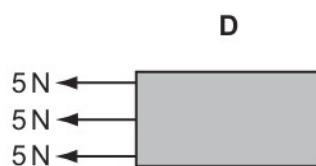
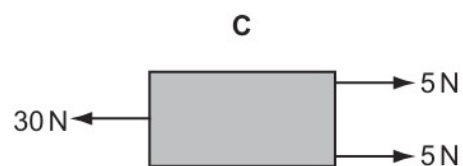
A car moves in a circle at constant speed.

What is the direction of the resultant force acting on the car?



73. O/N/15/11/Q4

Which object has the largest resultant force?



74. O/N/15/11/Q5

In which example does friction act in the direction of forward motion of the object on which it acts?

- A** on a suitcase accelerating on a conveyor belt
- B** on sandpaper rubbing a wooden block
- C** on the skis of a skier accelerating downhill on smooth snow
- D** on the tyres of a braking car

75. O/N/15/12/Q5

An apple of mass 0.15 kg and weight 1.5 N falls from a tree. At one point during its fall, the air resistance on the apple is 0.60 N upwards.

What is the acceleration of the apple at this point?

- A** 4.0 m/s²
- B** 6.0 m/s²
- C** 10 m/s²
- D** 14 m/s²

76. O/N/15/12/Q6

At time t_1 , a stone is dropped from a stationary balloon. The stone reaches terminal velocity at time t_2 .

Which row gives the acceleration of the stone at time t_1 and at time t_2 ?

	<u>acceleration at t_1</u> m/s ²	<u>acceleration at t_2</u> m/s ²
A	0	0
B	10	0
C	0	10
D	10	10

77. O/N/15/12/Q7

Four of the gravitational forces that act between bodies in the Solar System are described below.

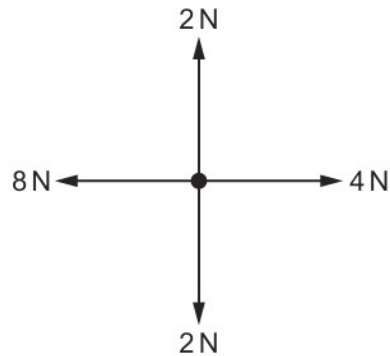
- P** the force on the Moon due to the Earth
- Q** the force on the Earth due to the Sun
- R** the force on the Earth due to the Moon
- S** the force on the Moon due to the Sun

Which two forces are a Newton's third law pair (action and reaction)?

- A** P and Q
- B** P and R
- C** Q and S
- D** R and S

78. M/J/15/11/Q6

Four forces act at a point as shown.



What is the size of the resultant force?

- A** 0 N **B** 4 N **C** 6 N **D** 8 N

79. M/J/15/11/Q9

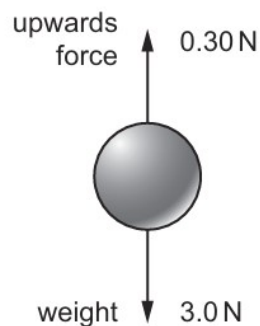
Newton's third law involves two quantities which are equal in size and opposite in direction.

What is the unit for these two quantities?

- A** J
B m/s^2
C N
D W

80. M/J/15/12/Q5

A metal ball of mass 0.30 kg and weight 3.0 N is held so that it is below the surface of oil. It experiences an upwards force of 0.30 N.



When the ball is released, what is its initial acceleration?

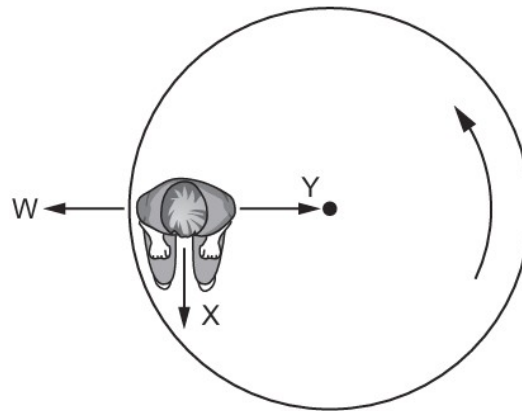
- A** 1.0 m/s^2 **B** 9.0 m/s^2 **C** 10 m/s^2 **D** 11 m/s^2

81. O/N/14/11/Q5

A boy sits on a playground roundabout (carousel).

The roundabout carries the boy in a horizontal, anticlockwise circle at a constant speed.

The diagram shows the view of the roundabout from above.



What describes the resultant force on the boy?

- A The resultant force is zero.
- B The resultant force is in direction W.
- C The resultant force is in direction X.
- D The resultant force is in direction Y.

82. O/N/14/12/Q1

When a heavy coin falls a short distance towards the ground it does **not** reach terminal velocity.

Why is this?

- A The coin has not hit the ground.
- B The weight of the coin equals the air resistance.
- C The weight of the coin increases as air resistance increases.
- D The weight of the coin is always more than air resistance.

83. O/N/14/12/Q2

A car travels along a road at 50 km/h.

The driver applies the same braking force at the same place on a day when the surface is dry and then on a day when the road is wet.

On the **wet** surface, how many of these distances are greater than on the dry surface?

braking distance

stopping distance

thinking distance

- A 0 B 1 C 2 D 3

84. M/J/14/11/Q4

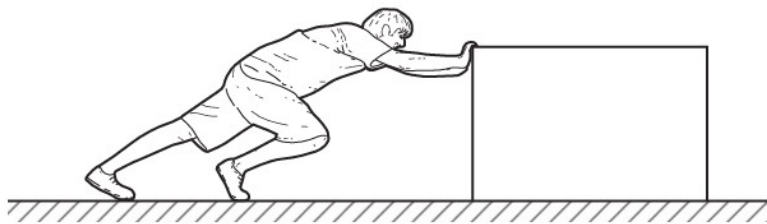
An object falls from rest through the air and the air resistance on it increases. The object reaches terminal velocity.

Which quantity **decreases** until its terminal velocity is reached?

- A acceleration
- B kinetic energy
- C speed
- D weight

85. M/J/14/11/Q5

A man pushes a heavy box along the ground.



A force acts between the man's hands and the box.

Another force acts between the man's feet and the floor.

In which direction do these forces act on the man?

	force on man's hands	force on man's feet
A	towards the left	towards the left
B	towards the left	towards the right
C	towards the right	towards the left
D	towards the right	towards the right

86. M/J/14/11/Q6

In which situation is the resultant force on the body equal to zero?

- A a car turning a corner at constant speed
- B a rock falling freely on the Moon
- C a train going up a straight hill at constant speed in a straight line
- D an aircraft accelerating along a runway in a straight line

87. M/J/14/11/Q7

A satellite is orbiting the Earth.

What is the direction of the force on the satellite causing this circular motion?

- A away from the centre of the Earth
- B in the direction of motion of the satellite
- C in the opposite direction to the motion of the satellite
- D towards the centre of the Earth

88. M/J/14/11/Q11

A force is applied to a body.

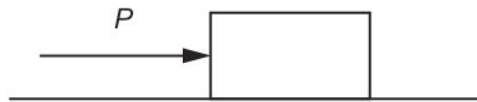
Which property of the body **cannot** be changed by the force?

- A its mass
- B its shape
- C its size
- D its velocity

89. M/J/14/12/Q5

The diagram shows a block of stone on a rough horizontal surface.

Force P acts on the block as shown.



The block is at rest. A frictional force F acts on the block.

Which row shows the direction and size of F ?

	direction of F	size of F
A	to the left	more than P
B	to the right	more than P
C	to the left	same as P
D	to the right	same as P

90. O/N/13/11/Q4

A student kicks a ball horizontally along the ground. As he does so, his foot applies a force to the ball.

At the same time, the ball applies a force to his foot.

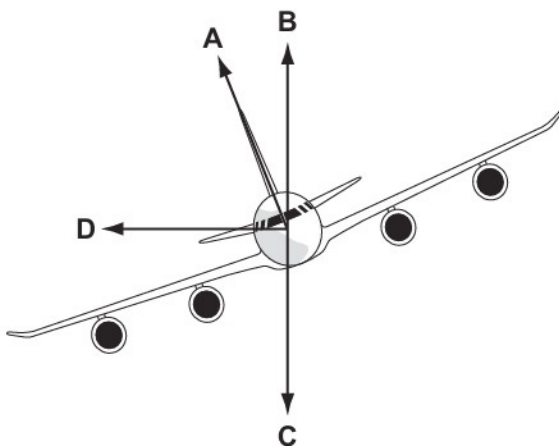
How do these forces compare?

- A** The force on the foot is equal to the force on the ball and in the opposite direction.
- B** The force on the foot is equal to the force on the ball and in the same direction.
- C** The force on the foot is smaller than the force on the ball and in the opposite direction.
- D** The force on the foot is smaller than the force on the ball and in the same direction.

91. O/N/13/11/Q5

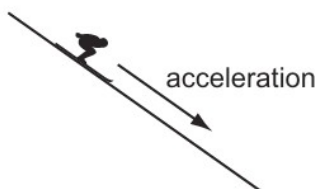
The diagram shows an aeroplane turning in a horizontal circle at constant speed.

In which direction is there a resultant force?



92. O/N/13/12/Q4

A skier of mass 80 kg accelerates down a slope at 2 m/s^2 .



The gravitational field strength g is 10 N/kg .

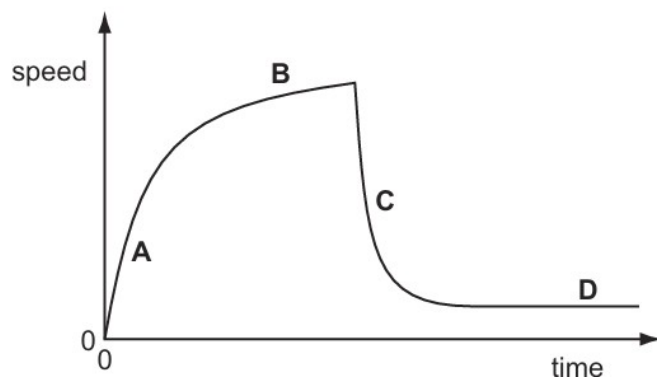
What is the weight of the skier and the resultant force on the skier?

	weight / N	resultant force / N
A	8	168
B	80	240
C	800	160
D	800	640

93. M/J/13/11/Q3

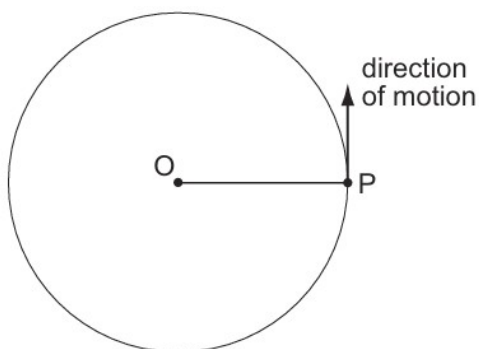
The speed-time graph for a falling skydiver is shown below. As he falls, the skydiver spreads out his arms and legs and then opens his parachute.

Which part of the graph shows the skydiver falling with terminal velocity?



94. M/J/13/11/Q4

A particle P is moving in a horizontal circle about O. P moves at constant speed.



Which statement is true?

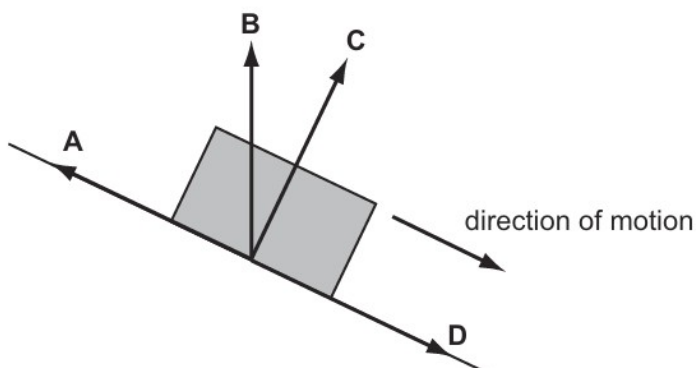
- A** A force of constant size acts on P in the direction of motion.
- B** A force of constant size acts on P towards O.
- C** The force on P varies in size as it moves around the circle.
- D** There is no resultant force acting on P.

95. M/J/13/11/Q5

A body slides down a frictionless slope as shown.

As the body presses on the surface, the surface pushes back on the body. The force of the surface on the body is sometimes called the reaction force.

In which direction does the reaction force act?

**96. M/J/13/12/Q6**

Balanced forces are acting on a moving body.

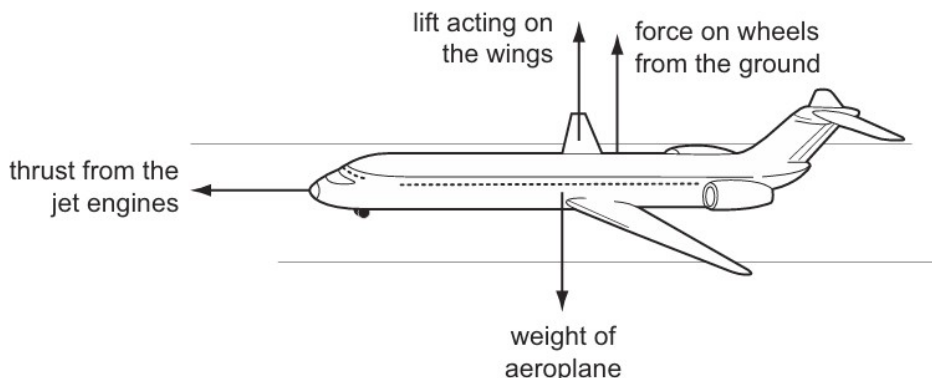
What happens to the direction of movement and to the speed of the body?

	direction of movement	speed
A	changes	changes
B	changes	does not change
C	does not change	changes
D	does not change	does not change

97. O/N/12/11/Q5

An aeroplane is half way along a runway before it takes off.

The arrows show the directions of four forces acting on the aeroplane.



The drag forces on the aeroplane are negligible.

The gravitational field strength g is 10 N/kg .

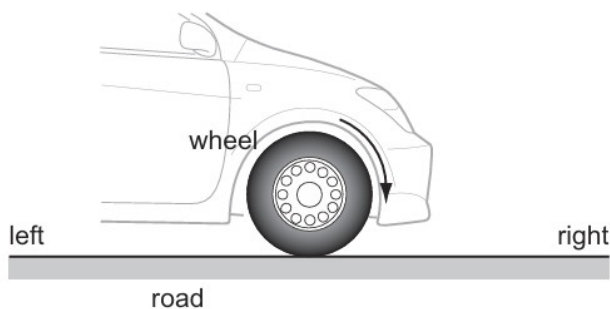
The acceleration of the aeroplane is 2 m/s^2 .

What is the largest force?

- A** the force on the wheels from the ground
- B** the lift acting on the wings
- C** the thrust from the jet engines
- D** the weight of the aeroplane

98. O/N/12/11/Q6

The front wheel of a car is turned in a clockwise direction by the engine as the car accelerates towards the right, as shown in the diagram. There is a force of friction between the wheel and the road.

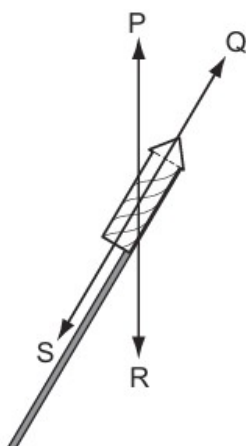


What are the directions of the frictional forces on the wheel of the car and on the road?

	direction of frictional force on wheel	direction of frictional force on road
A	to the left	to the left
B	to the left	to the right
C	to the right	to the left
D	to the right	to the right

99. O/N/12/12/Q5

The diagram shows a firework rocket.



As the rocket flies through the air, three forces act on it. These forces are weight, thrust and air resistance.

What are the three forces?

	thrust	air resistance	weight
A	P	R	S
B	P	S	R
C	Q	R	S
D	Q	S	R

100. O/N/12/12/Q6

An object has a mass of 15 kg. It is pushed horizontally by a force of 40 N. The frictional force is 10 N.

What is the acceleration of the object?

- A** 0.50 m/s^2 **B** 1.5 m/s^2 **C** 2.0 m/s^2 **D** 2.7 m/s^2

101. M/J/12/11/Q5

A hard stone hits the ground and comes to rest almost immediately.

As the stone hits the ground, what is the direction and the size of the force acting on the ground?

- A** downwards and equal to the weight of the stone
- B** downwards and larger than the weight of the stone
- C** upwards and equal to the weight of the stone
- D** upwards and larger than the weight of the stone

102. M/J/12/11/Q6

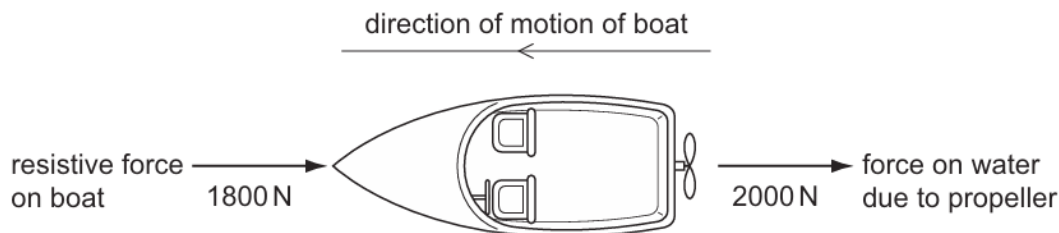
A car of mass 1500 kg is towing a trailer of mass 1100 kg along a level road. The acceleration of the car is 1.30 m/s^2 .

Ignoring friction and air resistance, what is the driving force on the car?

- A** 1430 N **B** 1950 N **C** 2000 N **D** 3380 N

103. O/N/11/11/Q4

The propeller on a boat pushes water backwards with a force of 2000 N. The boat moves through the water against a total resistive force of 1800 N.



According to Newton's third law, what is the forward force on the propeller due to the water?

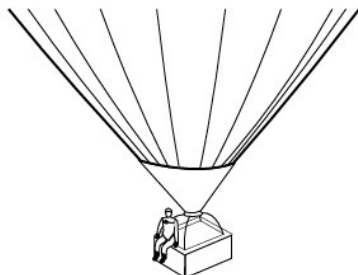
- A** 200 N **B** 1800 N **C** 2000 N **D** 3800 N

104. O/N/11/11/Q5

The diagrams show a parachutist in four positions after she jumps from a high balloon.

At which position does she have terminal velocity?

- A** not yet moving



- B** accelerating



- C** not accelerating



- D** not moving

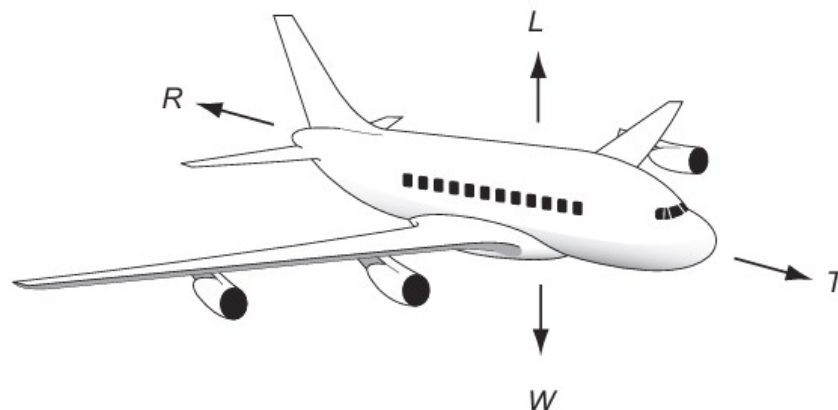


105. O/N/10/11/Q5

An aircraft, flying at a constant height, is gaining speed.

The four forces acting are

- L lift due to the wings
- R air resistance
- T the thrust due to the engines
- W the weight



Which row is correct?

	vertical forces	horizontal forces
A	$L = W$	$T = R$
B	$L > W$	$T > R$
C	$L = W$	$T > R$
D	$L > W$	$T = R$

106. O/N/10/11/Q6

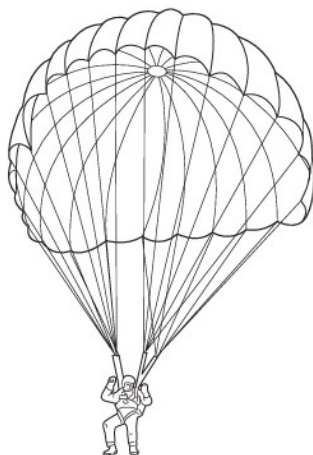
A car travels along a road. The driver stops the car by pushing his foot down on the brake pedal.

What does **not** change if he pushes harder on the brake pedal?

- A** the braking distance
- B** the braking force
- C** the stopping distance
- D** the thinking distance

107. M/J/10/11/Q5

Two men jump out of an aeroplane at the same time. One of the men opens his parachute and the other man remains in free-fall.



Why is the man in free-fall moving faster than the parachutist?

- A** The man in free-fall experiences greater air resistance.
- B** The man in free-fall has a greater mass.
- C** The parachutist experiences greater air resistance.
- D** The parachutist has not reached terminal velocity.

108. O/N/09/Q5

A tractor pulls a trailer at a constant speed.

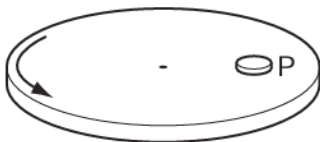
The tractor exerts a forward force of 1600 N on the trailer.

What is the force exerted by the trailer on the tractor?

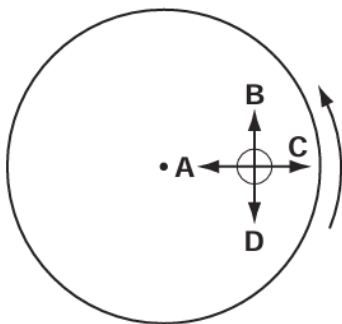
- A** 0 N
- B** 1600 N backwards
- C** 1600 N forwards
- D** 3200 N forwards

109. O/N/09/Q6

A turntable rotates at constant speed. A coin is placed on the turntable at P. The friction force between the coin and the turntable keeps the coin in the same position on the turntable.

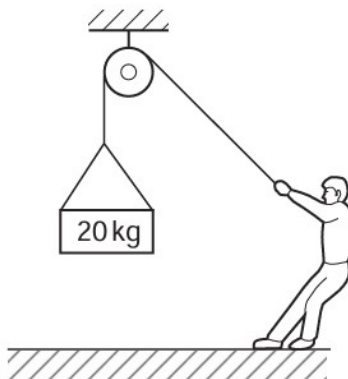


In which direction does the friction force act?



110. M/J/09/Q5

A person just supports a mass of 20 kg suspended from a rope.



What is the resultant force acting on the mass?

- A** 0 N **B** 10 N **C** 20 N **D** 200 N

111. M/J/08/Q5

What keeps an electron moving in a circle around the nucleus of an atom?

- A** a gravitational force away from the nucleus
- B** a gravitational force towards the nucleus
- C** an electrostatic force away from the nucleus
- D** an electrostatic force towards the nucleus

112. O/N/07/Q2

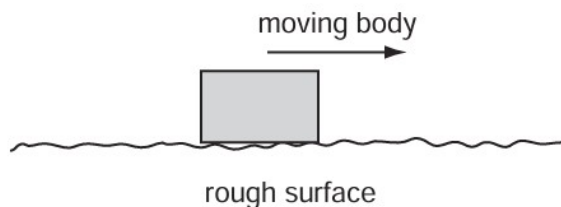
Forces of 30 N and 50 N act on the same body, but in different directions.

Which value could **not** be the resultant force on the body?

- A** 10 N **B** 30 N **C** 50 N **D** 70 N

113. O/N/07/Q4

When a body moves across a rough surface, a frictional force is produced.



Which statement about this force is **always** true?

- A** It acts in the direction of the motion.
B It is equal in value to the force producing the motion.
C It makes the body recoil in the opposite direction after stopping it.
D It opposes the motion across the surface.

114. O/N/06/Q4

When a block of wood of mass 2 kg is pushed along the horizontal flat surface of a bench, the friction force is 4 N.

When the block is pushed along the bench with a force of 10 N, it moves with a constant

- A** speed of 3 m/s.
B speed of 5 m/s.
C acceleration of 3 m/s².
D acceleration of 5 m/s².

115. M/J/06/Q2

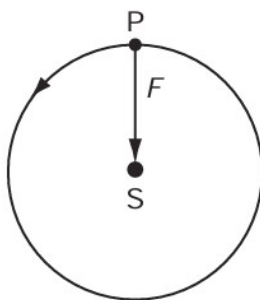
A force of 20 N pushes an object of mass 5.0 kg along a rough horizontal surface where the frictional force is 5.0 N.

What is the acceleration of the object?

- A** 1.0 m/s² **B** 2.0 m/s² **C** 3.0 m/s² **D** 4.0 m/s²

116. O/N/05/Q5

A body P moves in a circle around a point S . A force F keeps it moving in the circle.

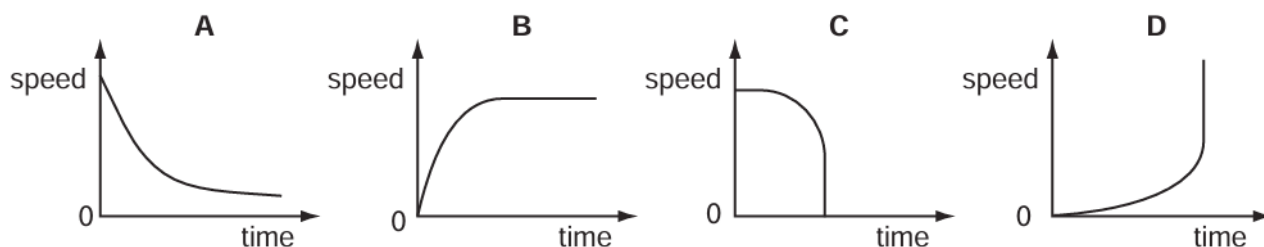


What happens if the force F suddenly disappears?

- A** P moves directly towards S .
- B** P moves in a circle closer to S .
- C** P moves away from S in a curved path.
- D** P goes off in a straight line.

117. M/J/05/Q2

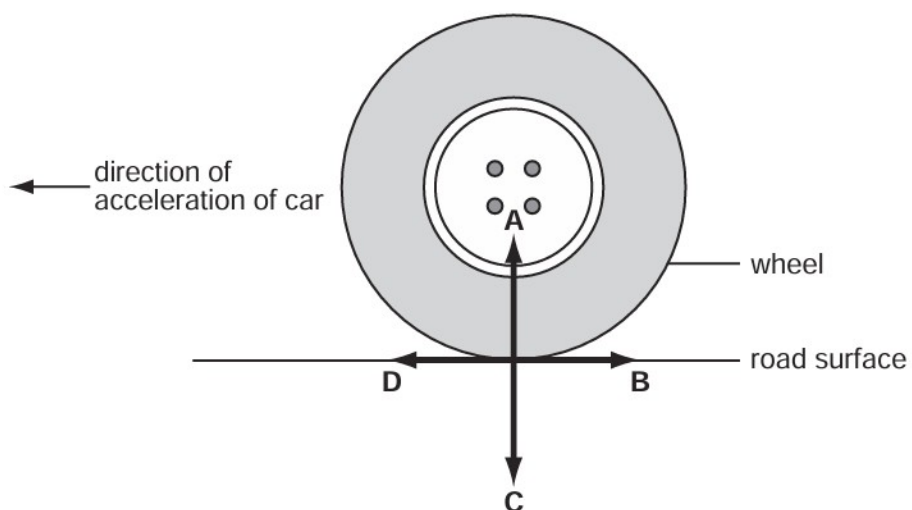
Which graph represents the motion of a body falling vertically that reaches a terminal velocity?



118. O/N/04/Q5

The wheel of a moving car is driven by the engine. The car is accelerating in the direction shown.

In which direction does the frictional force act on the wheel?



119. M/J/04/Q6

Which is a statement of Newton's third law of motion?

- A** Every force causes a reaction.
- B** If there is no resultant force on a body then there is no acceleration.
- C** The forces acting on a body are always equal and opposite.
- D** To every action there is an equal but opposite reaction.

ANSWERS

1. B	47. D	94. B
2. A	48. C	95. C
3. C	49. C	96. D
4. B	50. C	97. D
5. A	51. B	98. C
6. C	52. D	99. D
7. C	53. C	100. C
8. C	54. D	101. B
9. B	55. A	102. D
10. C	56. D	103. C
11. B	57. D	104. C
12. A	58. C	105. C
13. D	59. A	106. D
14. A	60. A	107. C
15. C	61. C	108. B
16. A	62. C	109. A
17. C	63. B	110. A
18. A	64. A	111. D
19. C	65. D	112. A
20. D	66. A	113. D
21. D	67. D	114. C
22. C	68. C	115. C
23. D	69. C	116. D
24. C	70. A	117. B
25. D	71. B	118. D
26. D	72. B	119. D
27. A	73. C	
28. A	74. A	
29. D	75. B	
30. B	76. B	
31. D	77. B	
32. D	78. B	
33. A	79. C	
34. B	80. B	
35. A	81. D	
36. C	82. D	
37. D	83. C	
38. A	84. A	
39. C	85. B	
40. B	86. C	
41. B	87. D	
42. D	88. A	
43. C	89. C	
44. B	90. A	
45. A	91. D	
46. D	92. C	
	93. D	

1. O/N/23/21/Q2

A lorry (truck) is at rest on a straight, horizontal road.

A constant driving force of $5.4 \times 10^4 \text{ N}$ acts on the lorry and it accelerates forwards.

(a) Define what is meant by 'acceleration'.

.....
.....
..... [2]

(b) As the lorry moves forwards, work is done on it by the driving force.

(i) State the equation that defines work done.

Refer to the direction of the force exerted.

.....
..... [1]

(ii) Calculate the work done on the lorry by the driving force when the lorry travels a distance of 280 m.

work done = J [2]

(iii) At one point on the road, the kinetic energy of the lorry is $3.2 \times 10^6 \text{ J}$ and its speed is 12 m/s.

Calculate the mass of the lorry.

mass = kg [3]

[Total: 8]

2. M/J/23/22/Q1

Fig. 1.1 shows the speed–time graph for a car travelling on a straight horizontal road.

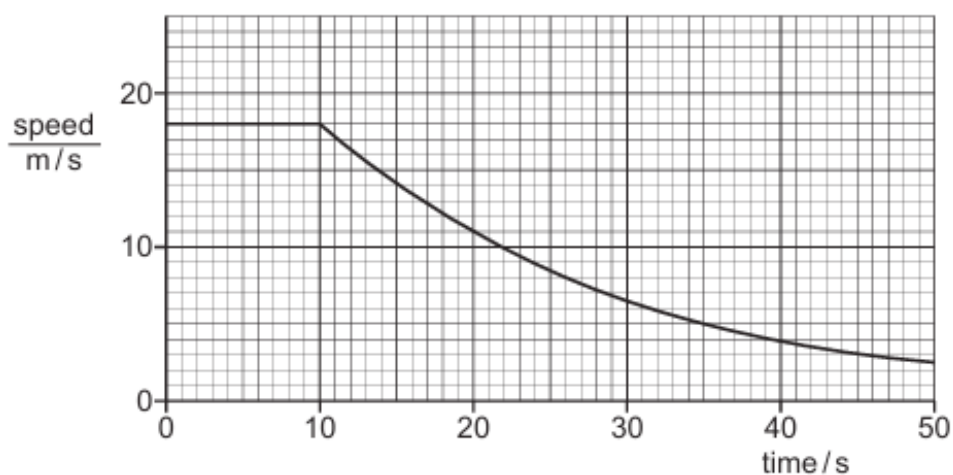


Fig. 1.1

- (a) Describe the motion of the car shown in Fig. 1.1.

.....

.....

.....

..... [2]

- (b) At time $t = 10$ s the engine of the car is switched off. The brakes are not applied.

- (i) Name **two** forces that act on the car to cause the change in motion after $t = 10$ s.

1

2

[1]

- (ii) Suggest why Fig. 1.1 is a curve after $t = 10$ s.

.....

.....

..... [1]

(c) Between $t = 10\text{ s}$ and $t = 20\text{ s}$ the speed of the car changes from 18 m/s to 11 m/s .

The mass of the car is 1200 kg .

(i) Calculate the change in momentum of the car in this time.

Give the unit of your answer

momentum change = unit [2]

(ii) Calculate the average resultant force exerted on the car during this time.

average resultant force = N [2]

[Total: 8]

3. O/N/22/21/Q7

Fig. 7.1 shows a child sitting on a sledge on a snow-covered hill of constant slope.

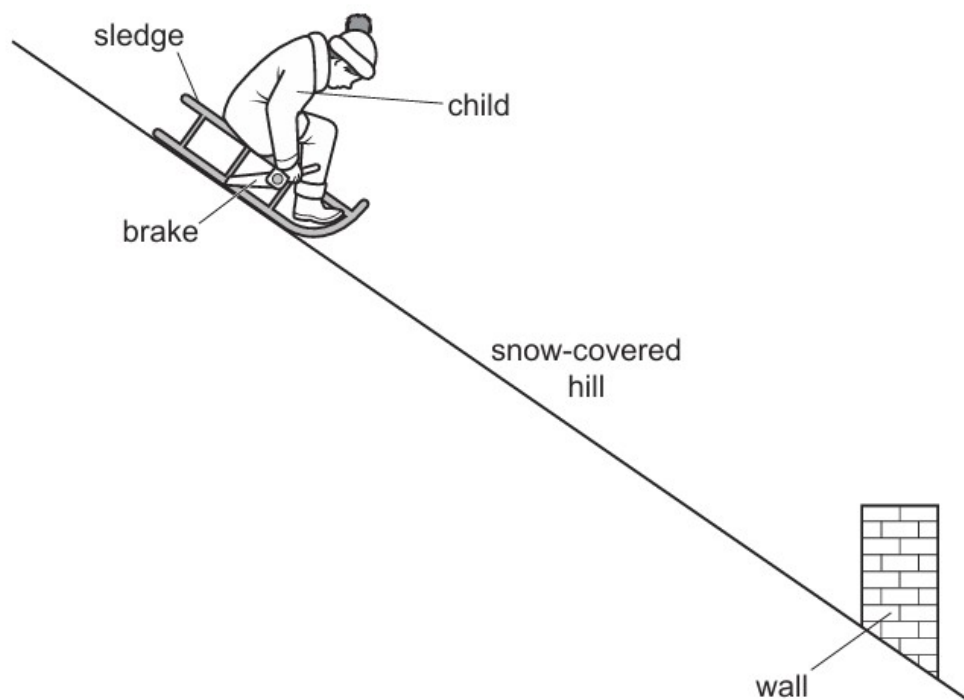


Fig. 7.1 (not to scale)

At time $t = 0$, the child and the sledge begin to move down the hill in a straight line.

When the child sees a wall ahead, he applies the brake.

The child and sledge continue to travel in a straight line until they come to a stop before hitting the wall.

Fig. 7.2 is the speed-time graph for the journey.

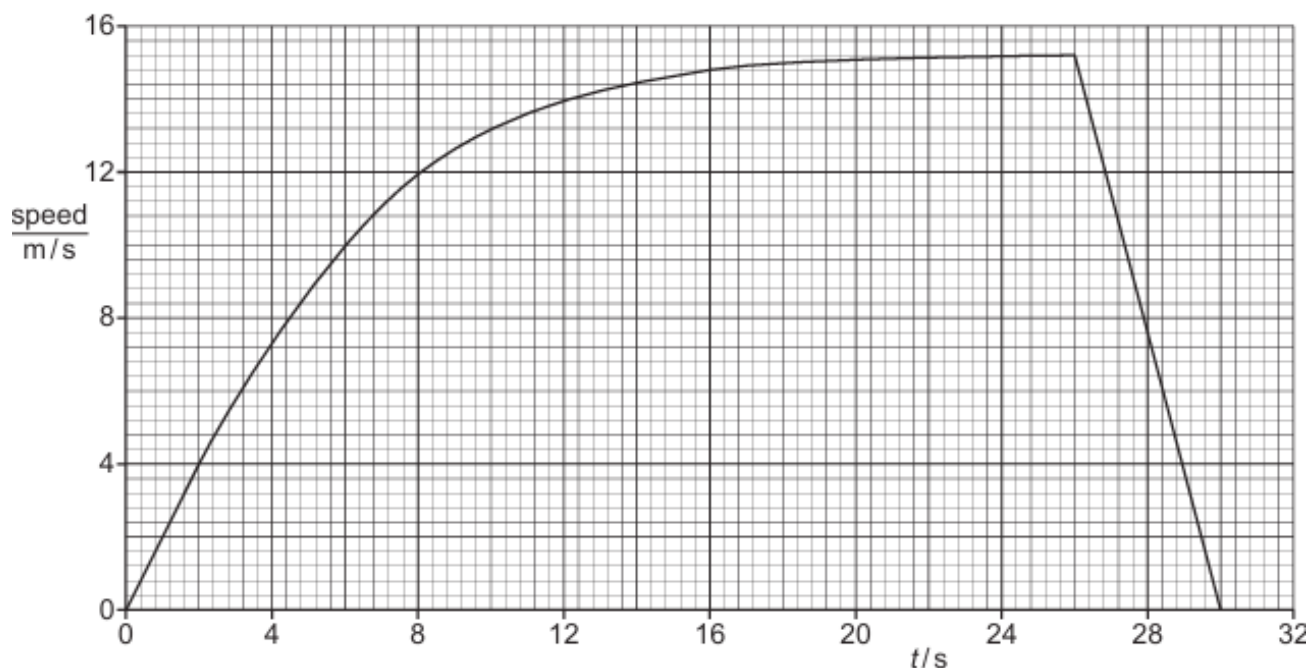


Fig. 7.2

The brake is applied at $t = 26$ s.

- (a) Fig 7.2 shows how the speed of the child and sledge varies over the whole of the journey.

Explain why, between $t = 0$ and $t = 26$ s, the speed varies in the way shown by the graph.

.....

.....

.....

.....

.....

..... [3]

- (b) At $t = 26$ s, the front of the sledge is 35 m from the wall.

Determine the distance between the front of the sledge and the wall when the sledge stops.

distance = [3]

(c) At $t = 26\text{ s}$, the child and sledge begin to decelerate.

(i) Determine the size of the deceleration.

deceleration = [3]

(ii) The mass of the child is 46 kg and the mass of the sledge is 9.0 kg .

Calculate the resultant force on the child and sledge as they decelerate.

resultant force = [2]

(iii) State the energy transfer that is taking place as the child and sledge decelerate.

.....
.....
..... [2]

(d) At $t = 26\text{ s}$, when the brake is first applied, the child jerks forwards on the sledge.

Explain why.

.....
.....
..... [2]

[Total: 15]

4. O/N/22/22/Q1

A train travels along a straight horizontal track. At time $t = 0$, the train passes through station P at constant speed without stopping.

The driver applies the brakes 70 s before reaching station Q. The train decelerates.

Fig. 1.1 is the speed–time graph for the train from $t = 0$ until it stops at station Q.

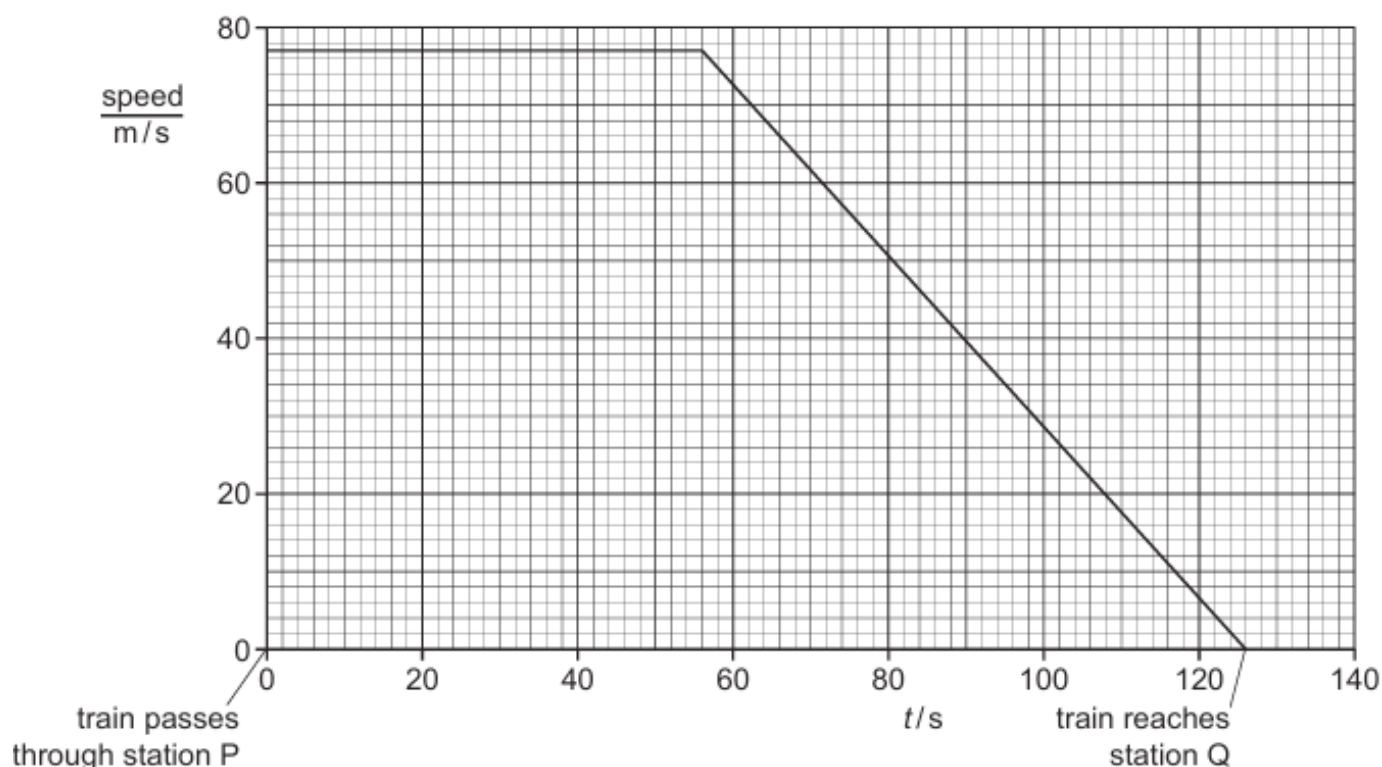


Fig. 1.1

(a) Using Fig. 1.1, determine the distance between station P and station Q.

distance = [3]

(b) The mass of the train is $3.8 \times 10^5 \text{ kg}$.

(i) Determine the deceleration of the train in the 70 s before it stops at station Q.

deceleration = [2]

(ii) Calculate the resultant force on the train as it decelerates.

resultant force = [2]

[Total: 7]

5. M/J/22/22/Q7

Fig. 7.1 shows the speed–time graph for a car travelling on a straight horizontal road.

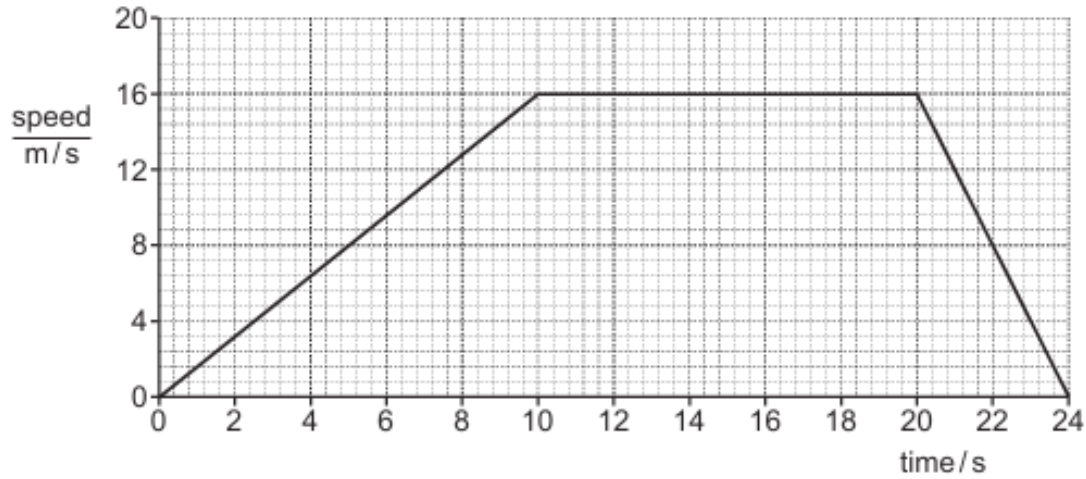


Fig. 7.1

- (a) (i) Describe the motion of the car.

.....
.....
.....
.....
..... [3]

- (ii) Using Fig. 7.1, calculate the distance travelled by the car during the 24 s of its motion.

Show your working.

distance = [3]

- (iii) Calculate the average speed of the car during its motion.

average speed = [2]

- (iv) A second car travels at a steady speed. It travels the same distance as the first car in the 24 s of the journey.

On Fig. 7.1, draw the speed–time graph for the second car. [2]

- (b) The thinking distance is the distance travelled by a car between the time that a hazard is seen and the time that the brakes are applied.

The braking distance is the distance travelled while the car slows down to rest.

Table 7.1 shows the thinking and braking distances for an alert driver when the car travels at different speeds.

Table 7.1

speed km/h	thinking distance / m	braking distance / m
20	9	2
40	18	9
60		20
80	36	36
100	45	56

- (i) Complete Table 7.1. [1]

- (ii) The time it takes for the driver to react to the hazard is constant at different speeds.

Explain how the table shows this.

.....

 [2]

- (iii) State what happens to the thinking distance and the braking distance when the driver is tired.

thinking distance
 braking distance [2]

[Total: 15]

6. M/J/22/21/Q7(c)

Fig. 7.1 shows a toy helicopter. It can hover and travel through the air.

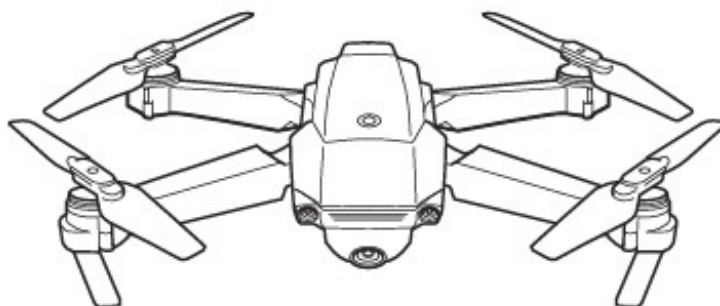


Fig. 7.1

A student flies the toy helicopter on a journey from A to B to C to D at a constant height.

Fig. 7.2 is a scale drawing of the path of the helicopter, viewed from above.

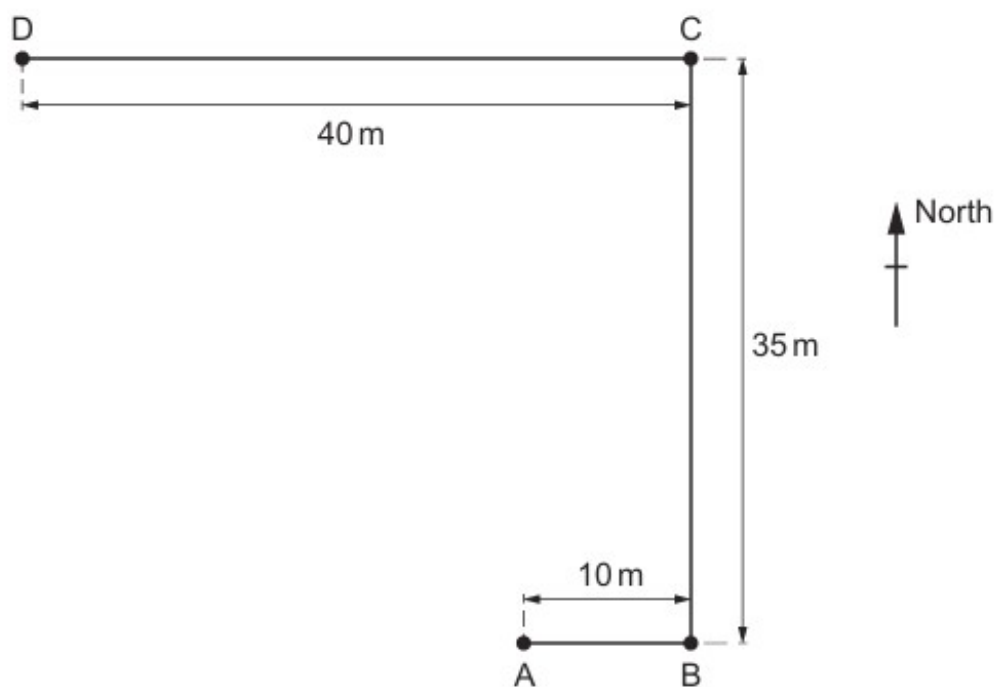


Fig. 7.2 (to scale)

- (c) When the toy helicopter hovers at D, its motor fails and it falls. It reaches terminal velocity as it falls.

Explain, in terms of the forces and acceleration, what happens as the helicopter falls and reaches terminal velocity.

.....

.....

.....

.....

.....

.....

.....

.....

..... [5]

7. O/N/21/21/Q2

Fig. 2.1 shows a man of mass 80 kg standing in a lift (elevator).

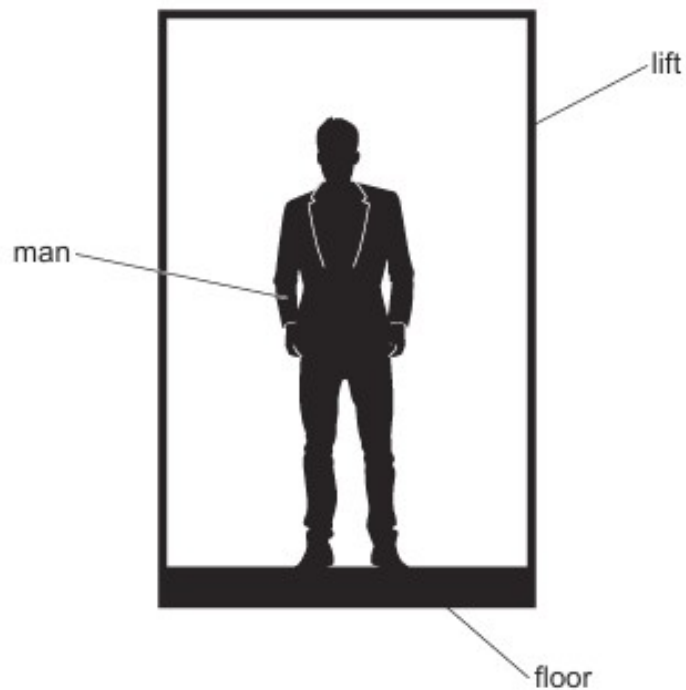


Fig. 2.1

The gravitational field strength g is 10 N/kg .

(a) Calculate the weight of the man.

weight = [1]

(b) The lift accelerates upwards uniformly at 0.50 m/s^2 .

Calculate:

(i) the resultant upward force on the man

resultant force = [2]

(ii) the force exerted on the man by the floor of the lift.

force = [1]

- (c) The lift accelerates upwards uniformly from rest for 4.0 s and then decelerates uniformly to rest in 4.0 s. The total distance travelled is 8.0 m.

On Fig. 2.2, sketch the distance-time graph for this journey.

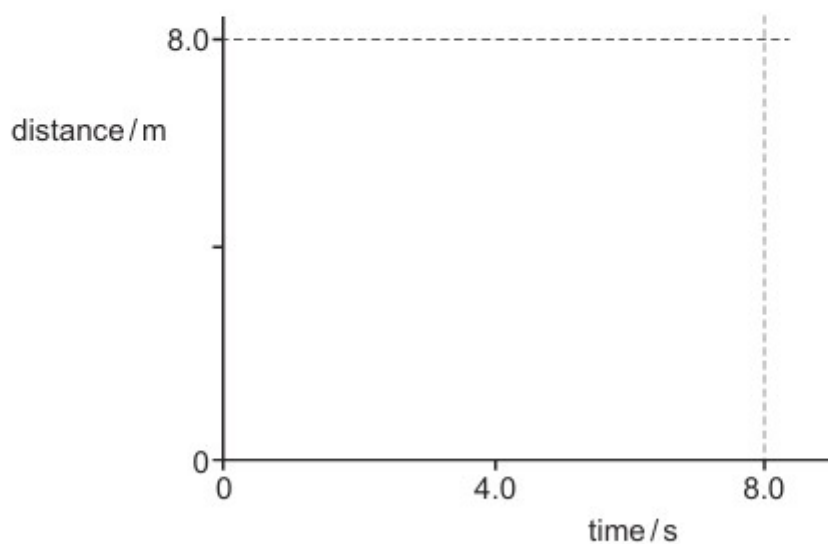


Fig. 2.2

[3]

[Total: 7]

8. O/N/21/22/Q2

Force is a vector quantity.

- (a) State what is meant by a *vector*.

.....
..... [1]

- (b) A swimmer reaches the end wall of a swimming pool and turns around under the water.

Fig. 2.1 shows the swimmer immediately after turning around.

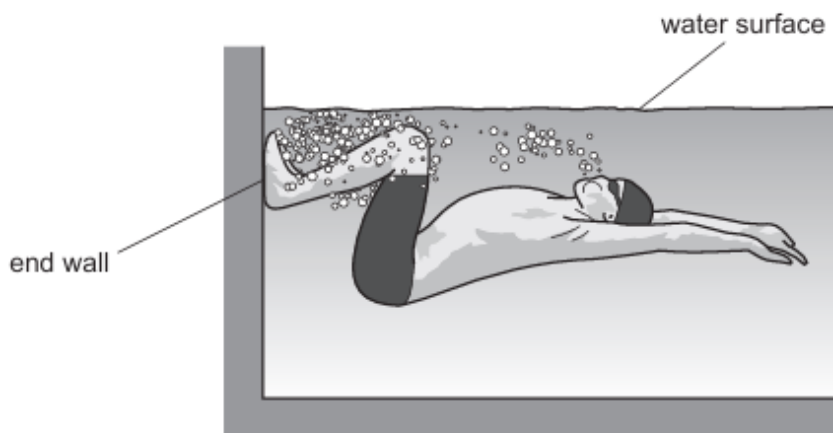


Fig. 2.1

- (i) The swimmer pushes against the end wall of the pool with his legs.

Explain, in terms of Newton's third law, why the swimmer accelerates away from the end wall.

.....
.....
.....
..... [3]

- (ii) While swimming, there is a constant forward force on the swimmer. His speed increases until eventually he reaches a constant speed.

Explain why he reaches a constant speed.

.....
.....
.....
..... [3]

[Total: 7]

9. M/J/21/21/Q8

Fig. 8.1 shows a stationary horse and its rider, about to jump over two fences.

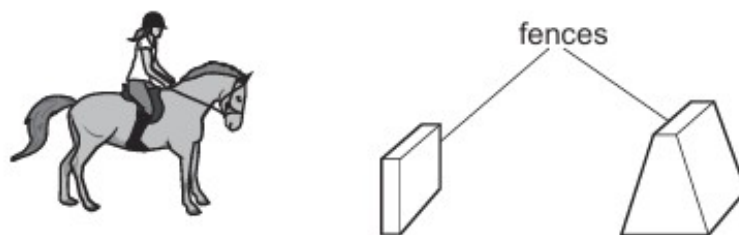


Fig. 8.1

(a) Fig. 8.2 shows a side view of the horse.

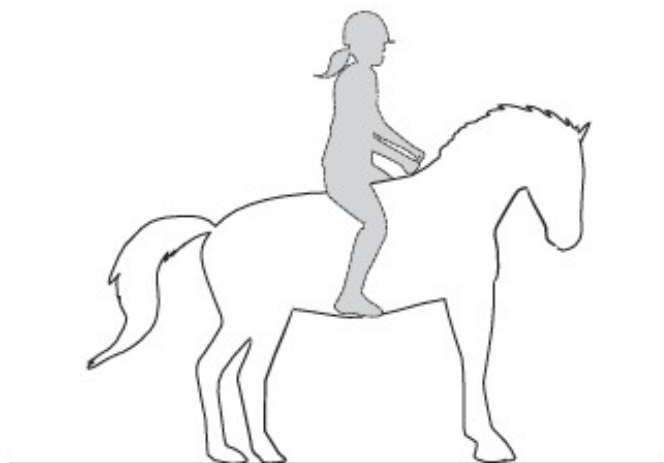


Fig. 8.2

(i) On Fig. 8.2, draw and label the forces acting on the horse.

Include the force that the rider exerts on the horse. Label this force F .

[3]

(ii) Explain how Newton's third law applies to force F .

.....
.....
..... [2]

10. O/N/20/21/Q1

Fig. 1.1 is the distance–time graph for a skydiver who jumps from a balloon at time $t = 0$.

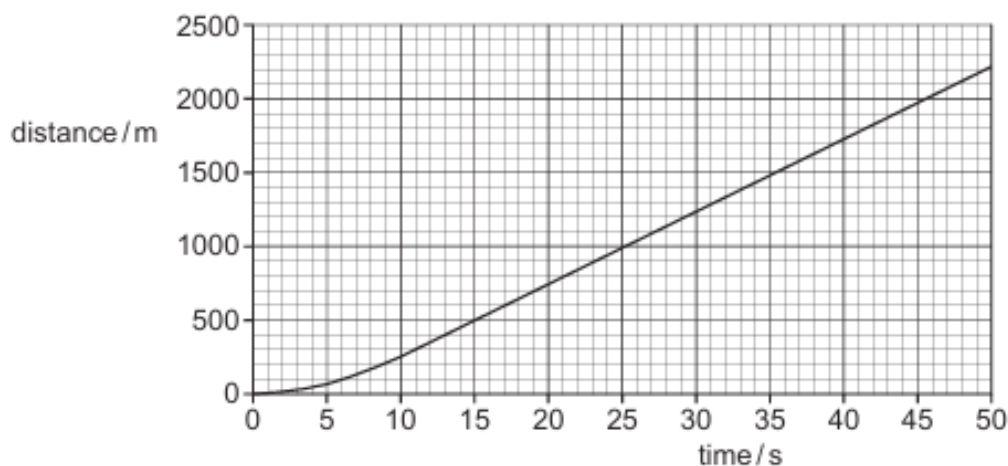


Fig. 1.1

- (a) The first part of the graph shows the motion of the skydiver from when he jumps until he reaches terminal velocity.

- (i) Describe the motion of the skydiver between $t = 0$ and $t = 20$ s.

.....

 [2]

- (ii) Explain the motion of the skydiver between $t = 0$ and $t = 20$ s in terms of the forces acting on him.

.....

 [3]

- (b) Using Fig. 1.1, determine the terminal velocity of the skydiver.

On Fig. 1.1, indicate any values used for your calculation.

terminal velocity = [3]

[Total: 8]

11. O/N/20/21/Q2

Fig. 2.1 shows a satellite moving at a constant speed in a circular orbit around the Earth.

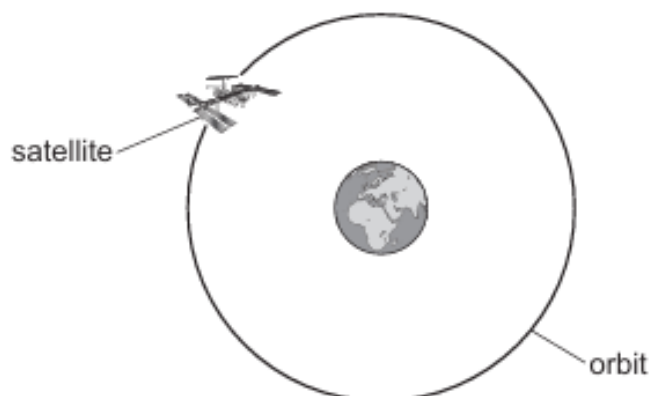


Fig. 2.1 (not to scale)

Speed is a scalar quantity but velocity is a vector quantity.

- (a) State how a scalar quantity differs from a vector quantity.

.....
..... [1]

- (b) Underline every vector quantity in the list.

distance displacement force length mass time
..... [1]

- (c) There is a resultant force acting on the satellite in Fig. 2.1.

- (i) Explain how the motion of the satellite shows that a resultant force is acting on it.

.....
.....
..... [2]

- (ii) State the cause of this force.

.....
..... [1]

[Total: 5]

12. O/N/20/22/Q7

A bus leaves a bus-stop at time $t = 0$ and travels along a horizontal road until it reaches a second bus-stop. Fig. 7.1 is the distance-time graph for the bus between $t = 0$ and $t = 60$ s.

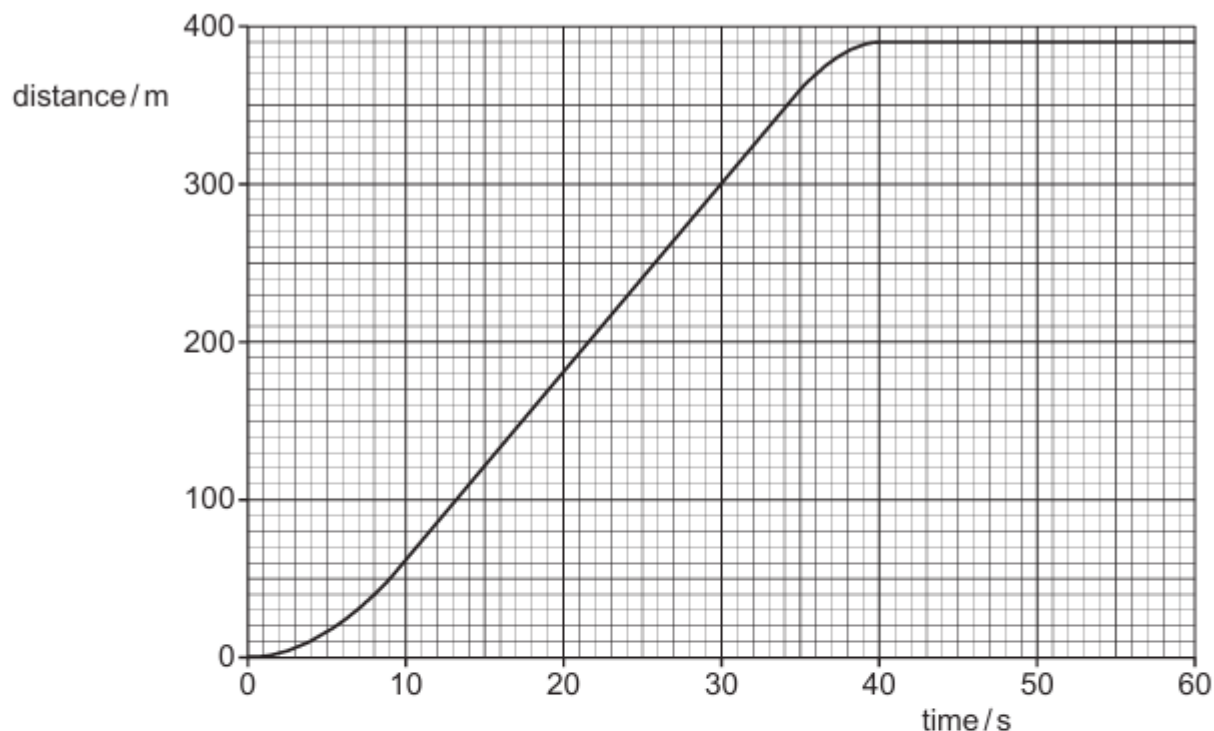


Fig. 7.1

The road on which the bus is travelling is straight except for a short, curved section. The bus travels around this circular curve between $t = 21$ s and $t = 24$ s.

- (a) Describe how the motion of the bus between $t = 0$ and $t = 10$ s differs from its motion between $t = 35$ s and $t = 40$ s.

.....

.....

.....

.....

..... [3]

(b) Determine:

- (i) the maximum speed of the bus during these 60 s

maximum speed = [3]

- (ii) the average speed of the bus between leaving the first bus-stop and arriving at the second bus-stop.

average speed = [2]

(c) (i) State how *velocity* differs from *speed*.

.....
..... [1]

- (ii) There are **three** periods during the 60 s when there is a non-zero resultant force acting on the bus.

Complete the statements to indicate these three time periods and state the direction of the resultant force in that period.

1. Between $t = \dots\dots\dots$ and $t = \dots\dots\dots$ the direction of the resultant force is
.....

2. Between $t = \dots\dots\dots$ and $t = \dots\dots\dots$ the direction of the resultant force is
.....

3. Between $t = \dots\dots\dots$ and $t = \dots\dots\dots$ the direction of the resultant force is
.....

[4]

(d) During the journey, the air resistance acting on the bus varies.

- (i) State why the air resistance changes during the journey.

.....
..... [1]

- (ii) On Fig. 7.1, mark and label with an M a time when the air resistance is a maximum value.
[1]

13. M/J/20/22/Q2

Fig. 2.1 shows the thinking distance and the braking distance for a car driven at 100 km/h.

The car has old, smooth tyres.

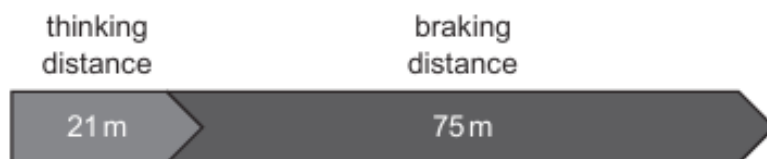


Fig. 2.1

- (a) Calculate the total stopping distance for the car.

stopping distance = [1]

- (b) The car is now fitted with new tyres.

At a speed greater than 100 km/h, the total stopping distance is the same as in (a).

- (i) State and explain the effect that the increase in speed and the use of new tyres have on the thinking distance.

effect

explanation

.....

..... [2]

- (ii) State and explain the effect that the increase in speed and the new tyres have on the braking distance.

effect

explanation

.....

..... [2]

[Total: 5]

14. M/J/20/21/Q1

Fig. 1.1 shows the thinking distance and the braking distance for a car being driven along a dry road and along a wet road at the same speed.

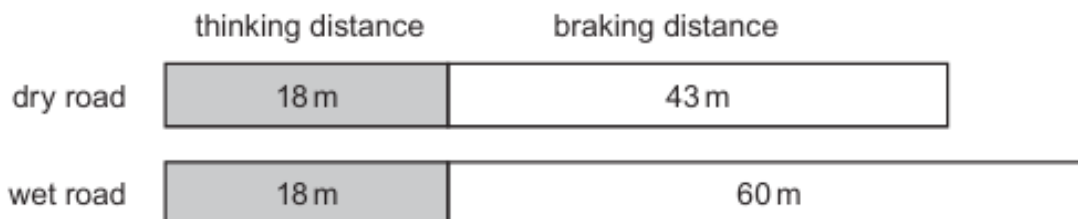


Fig. 1.1

- (a) Calculate the total stopping distance for the car on the wet road.

distance = [1]

- (b) Complete the sentence.

The thinking distance is the distance travelled between seeing a hazard and

..... [1]

- (c) (i) Suggest why the thinking distance is the same on both roads.

.....
.....
..... [1]

- (ii) Explain why the braking distance is larger when the road is wet.

.....
.....
.....
..... [2]

[Total: 5]

15. M/J/19/22/Q9(b)

Fig. 9.1 shows a map of the route taken by a car as it travels from town A to town B.

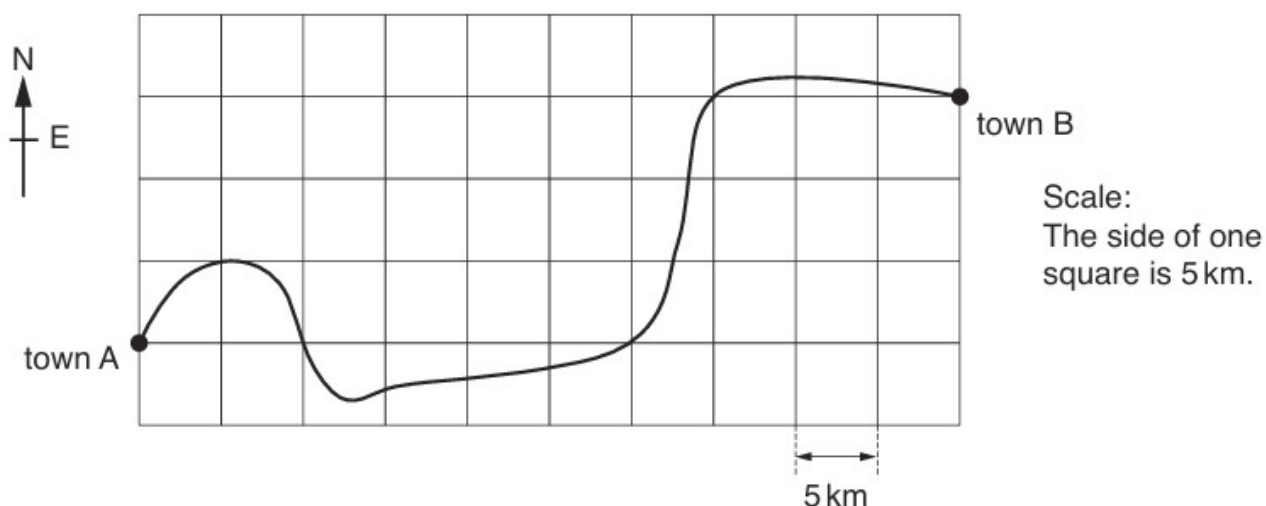


Fig. 9.1

- (b) At town B the car makes an emergency stop.

The total stopping distance of the car is the thinking distance added to the braking distance.

- (i) Describe what is meant by the *thinking distance*.

.....
 [1]

- (ii) The brakes on the car work badly. This increases the braking distance.

State **two** other factors that increase the braking distance.

1.

 2.

[2]

16. M/J/19/21/Q9

Fig. 9.1 shows a parachutist falling vertically towards the ground.



Fig. 9.1

The mass of the parachutist is 60 kg and his weight is 600 N.

(a) State what is meant by:

(i) *mass*

.....
..... [1]

(ii) *weight*.

.....
..... [1]

(b) Calculate the acceleration of the parachutist when the air resistance is 200 N.

acceleration = [3]

(c) The parachutist falls from rest at time $t = 0$.

He accelerates non-uniformly until he reaches a terminal velocity of 44 m/s at $t = 30$ s.

At $t = 40$ s the parachute opens. He decelerates for 4.0 s, reaching a smaller terminal velocity of 5.0 m/s.

He then falls at this speed until he lands on the ground at $t = 200$ s.

- (i) On Fig. 9.2, draw the speed–time graph for the parachutist.

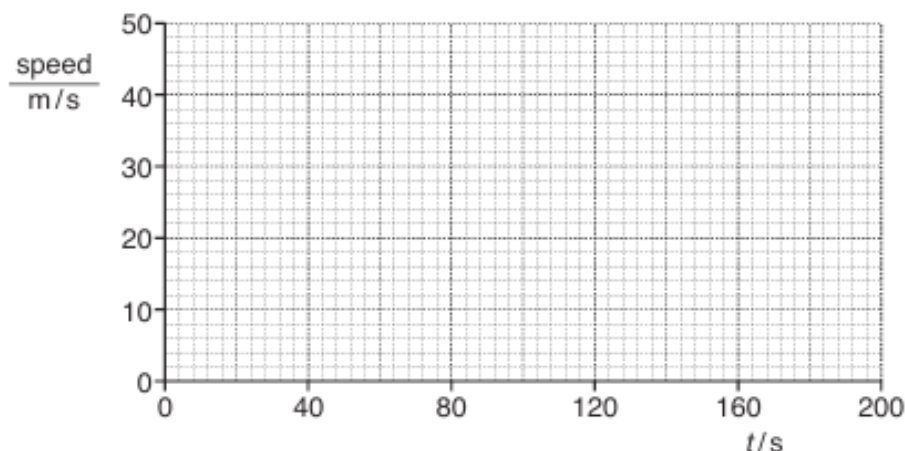


Fig. 9.2

[4]

- (ii) Calculate the average deceleration of the parachutist between $t = 40$ s and $t = 44$ s.

deceleration = [2]

- (d) Explain, in terms of the forces involved, why:

- (i) the parachutist reaches a terminal velocity at $t = 30$ s

.....
 [1]

- (ii) the parachutist decelerates when the parachute opens

.....

 [2]

- (iii) the second terminal velocity, when the parachute is open, is smaller than the first.

.....
 [1]

17. O/N/18/21/Q1

A girl of mass 35 kg, on a bicycle, accelerates from rest and travels down a slope in a straight line. The girl does not use the pedals.

Fig. 1.1 shows that the gradient of the slope is constant.

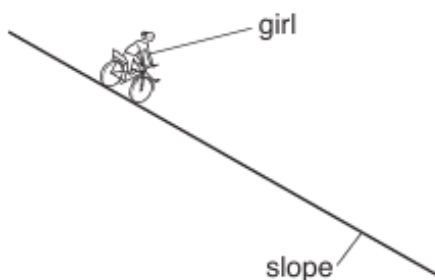


Fig. 1.1

- (a) Calculate the resultant force on the girl when she is accelerating at 2.6 m/s^2 .

resultant force = [2]

- (b) At first, her acceleration is constant. At time t_1 , her acceleration starts to decrease gradually until she is travelling at a constant speed in a straight line.

- (i) On Fig. 1.2, sketch a speed-time graph for the girl from when she starts moving until she is travelling at a constant speed. [2]

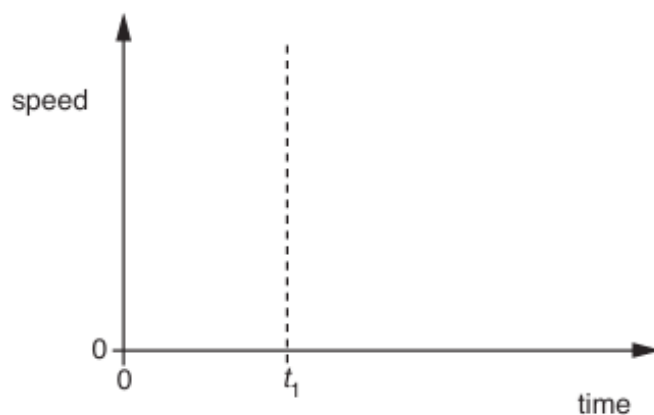


Fig. 1.2

- (ii) State how the distance travelled is found from a speed-time graph.

.....
..... [1]

18. M/J/18/21/Q2

Fig. 2.1 shows a man pushing a heavy box with a force P . A frictional force F acts in a horizontal direction.

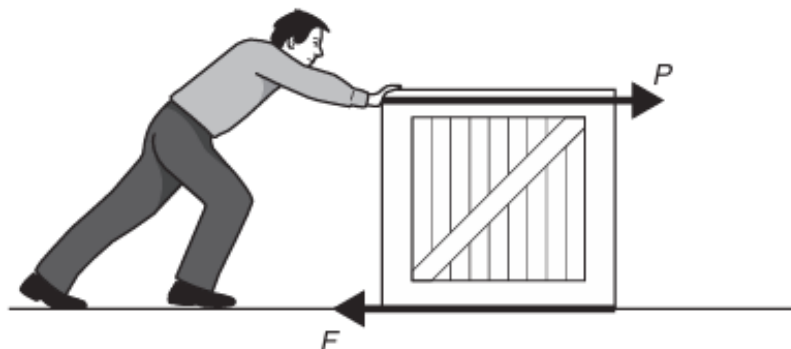


Fig. 2.1

(a) The forces on the box are *balanced* and the box is stationary.

(i) State what is meant by *balanced forces*.

.....
[1]

(ii) The frictional force in Fig. 2.1 does not produce any heating effect.

State what must happen for the frictional force to produce heating.

.....
[1]

(iii) Apart from being stationary, describe one other possible state of motion of the box when the forces are balanced.

.....
[1]

(b) When $P = 100\text{ N}$ and $F = 85\text{ N}$, the box accelerates. The mass of the box is 25 kg .

Calculate the acceleration of the box.

acceleration =[3]

19. M/J/17/21/Q1

Fig. 1.1 shows the directions of four forces acting on a racing car as it travels in a horizontal straight line.

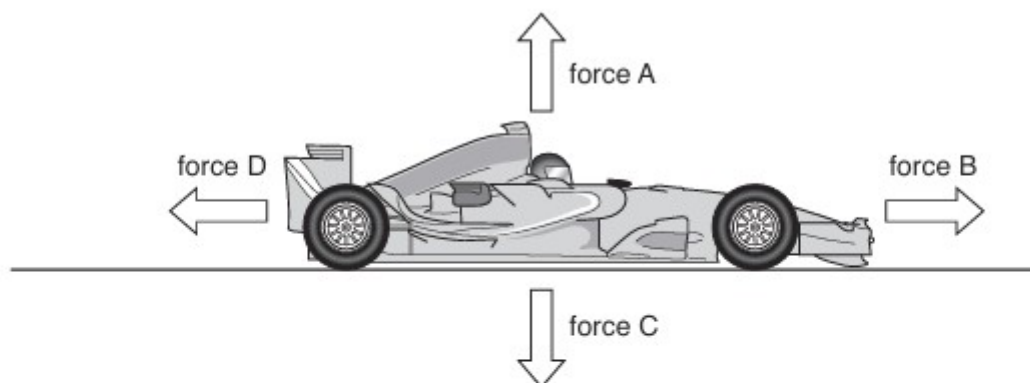


Fig. 1.1

(a) Draw a line from each box on the left to the correct description of each force.

force A	driving force
force B	contact or normal reaction force
force C	air resistance and friction
force D	force of gravity

[1]

(b) The table shows the sizes of the forces acting on the car at one time.

force A/N	force B/N	force C/N	force D/N
8000	1000	8000	600

The gravitational field strength g is 10 N/kg .

Calculate

(i) the mass of the car,

mass =[1]

(ii) the resultant force on the car,

resultant force =[1]

(iii) the acceleration of the car.

acceleration =[2]

(c) At another time, the car is travelling at speed u . It then accelerates for 5.0 s with an acceleration of 1.6 m/s^2 , and reaches a speed of 20 m/s.

Calculate the value of u .

$u = \dots\dots\dots$ [2]

20. M/J/16/21/Q12

The overall stopping distance of a cyclist is made up of two parts, as shown in Fig. 12.1:

- the distance the cyclist travels during the reaction time of the cyclist (the thinking distance)
- the distance the cyclist travels after the brakes are applied (the braking distance)

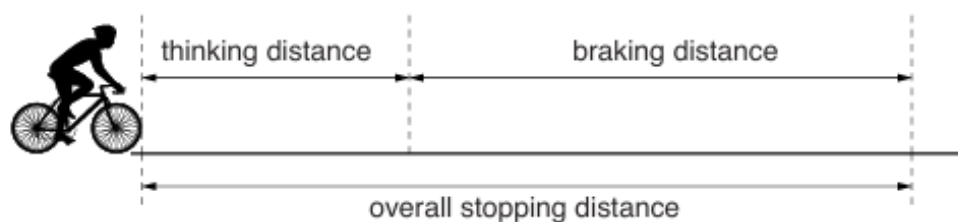


Fig. 12.1 (not to scale)

(a) State the energy change that occurs during braking.

.....
[2]

(b) A ball rolls in front of a cyclist at time $t = 0$ and the cyclist brakes and comes to rest.

Fig. 12.2 shows the speed-time graph for the cyclist.

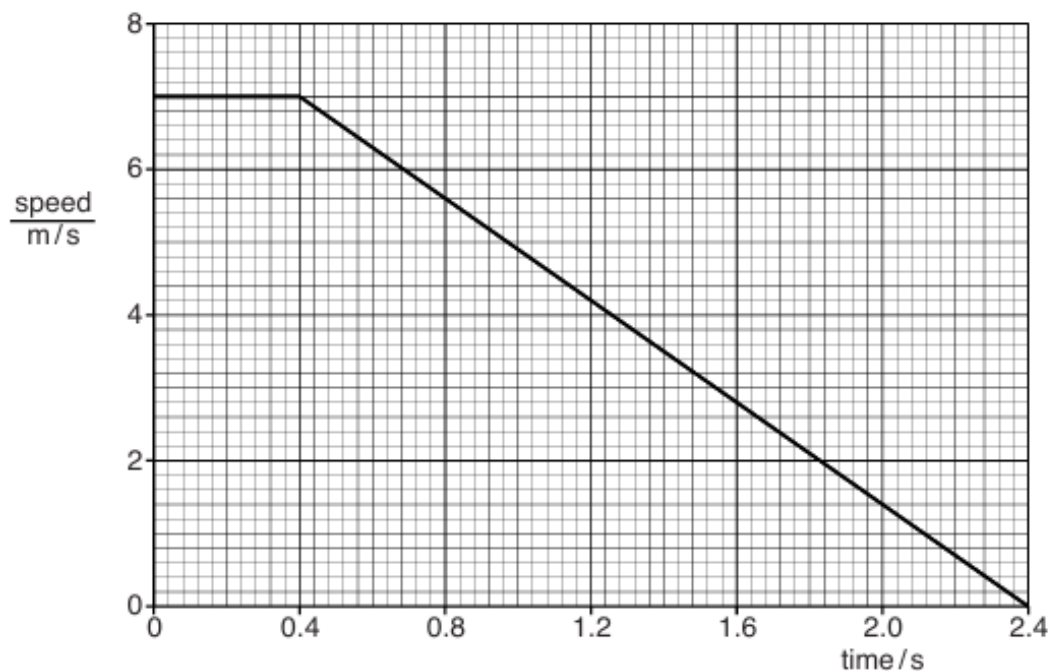


Fig. 12.2

(i) Using Fig. 12.2, determine the reaction time of the cyclist.

reaction time =[1]

- (ii) Using Fig. 12.2, calculate the thinking distance.

thinking distance =[2]

- (iii) State how the braking distance is found using Fig. 12.2.

.....
.....[1]

- (iv) On another occasion, the same cyclist travels at an initial speed of 5.0 m/s. A ball rolls in front of the cycle at time $t = 0$.

The cyclist has the same reaction time and the deceleration of the cycle is the same as in Fig. 12.2.

On Fig. 12.2 draw the new speed-time graph for the cyclist. [2]

- (v) The braking distance is longer when the cyclist stops on a wet road.

Explain why.

.....
.....
.....
.....
.....
.....[3]

21. M./J/14/22/Q1

Fig. 1.1 shows a lorry accelerating in a straight line along a horizontal road.



Fig. 1.1

- (a) The driving force on the lorry in the forward direction is D and the total backward force on the lorry is B .

(i) State and explain whether D or B is the larger force.

.....
..... [1]

(ii) Suggest one possible cause of the backward force B .

..... [1]

- (b) The weight of the lorry is 300 000 N.

The gravitational field strength g is 10 N/kg.

(i) Calculate the mass of the lorry.

mass = [1]

(ii) The resultant force on the lorry is 15 000 N. Calculate the acceleration of the lorry.

acceleration = [2]

- (c) Later, the lorry turns a corner at constant speed.

Explain why the lorry accelerates even though the speed is constant.

.....
..... [1]

22. O/N/11/22/Q2

Fig. 2.1 shows a skier of mass 85 kg skiing down a very steep slope.

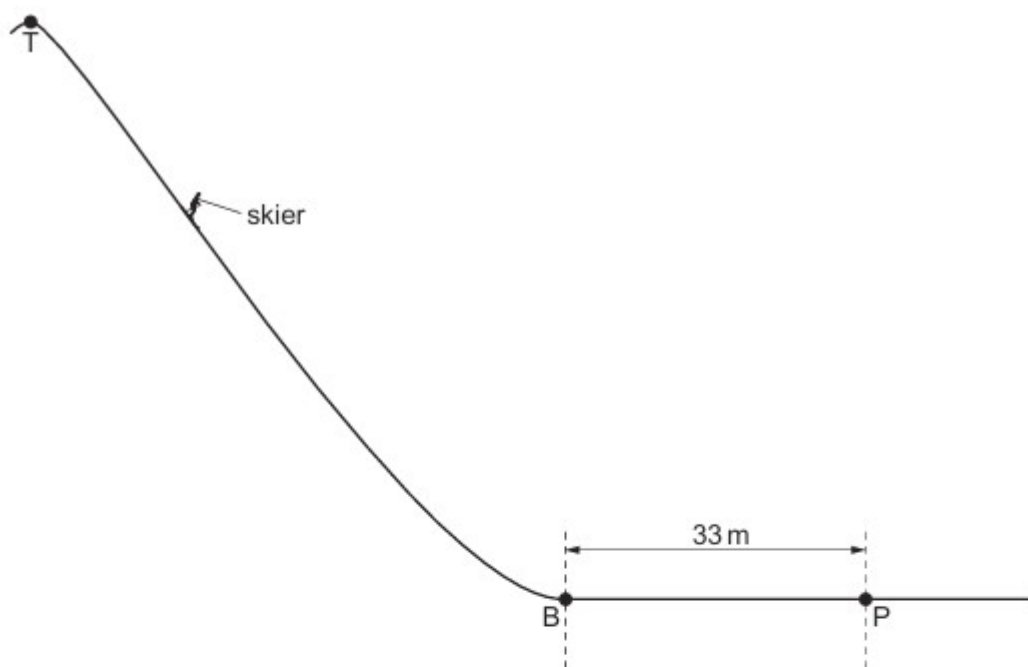


Fig. 2.1 (not to scale)

The skier starts from rest at the top T of the slope. The force of gravity accelerates him down the slope.

When he reaches the bottom B of the slope, his kinetic energy is $5.5 \times 10^4 \text{ J}$.

(a) The gravitational field strength is 10 N/kg . Calculate

(i) the weight of the skier,

weight =[1]

(ii) the minimum possible difference in height between T and B.

height difference =[2]

(b) At B, the skier digs his skis into the snow and stops at the point P after travelling 33 m horizontally.

Calculate the average horizontal force that acts on the skier between B and P.

force =[2]

23. M/J/15/21/Q9

Fig. 9.1 shows a satellite in orbit around the Earth.

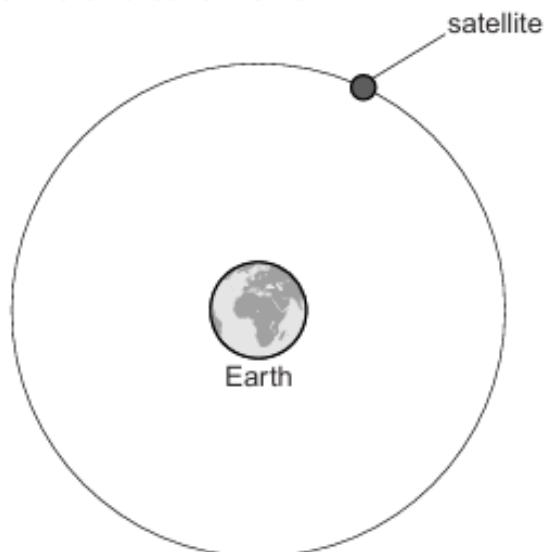


Fig. 9.1 (not to scale)

(a) The satellite travels at a constant speed in a circular orbit.

(i) Underline the quantities in the list below that are scalars.

acceleration force mass speed velocity [2]

(ii) The velocity of the satellite changes, but its speed is constant.

1. State what is meant by *velocity*.

.....
.....[1]

2. Explain why the velocity changes.

.....
.....[1]

(iii) Explain what makes this satellite move in an orbit that is circular.

.....
.....
.....[2]

- (b) The satellite is placed into orbit by a rocket. Fig. 9.2 shows the rocket as it takes off.

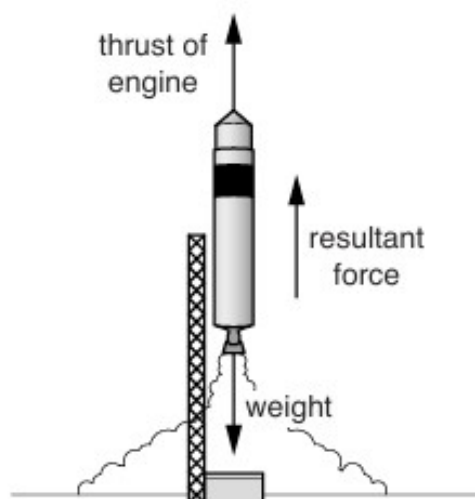


Fig. 9.2

The rocket and fuel have a total mass of 40 000 kg and a total weight of 400 000 N. The resultant force acting upwards on the rocket is 50 000 N.

- (i) Calculate the thrust produced by the rocket engine.

thrust =[1]

- (ii) Calculate the acceleration of the rocket.

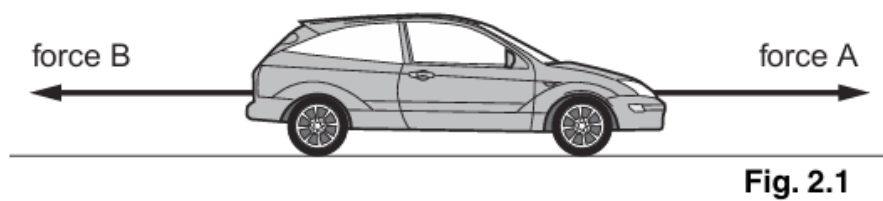
acceleration =[2]

24. M/J/12/22/Q2

(a) State what is meant by *friction*.

.....
.....[2]

(b) Fig. 2.1 shows two horizontal forces that act on a car. Force B is caused by air resistance and friction.



The car is travelling along a straight level road.

(ii) The mass of the car is 800 kg.
Force A increases to 5000 N. This causes the car to accelerate initially at 1.5 m/s^2 .
Calculate the size of force B.

force B =[3]

(iii) Force A remains constant at 5000 N. Explain why the acceleration decreases as the car travels along the level road.

.....
.....[2]

25. O/N/11/21/Q9

Fig. 9.1 is the speed-time graph for a rocket from the moment that the fuel starts to burn at time $t = 0$.

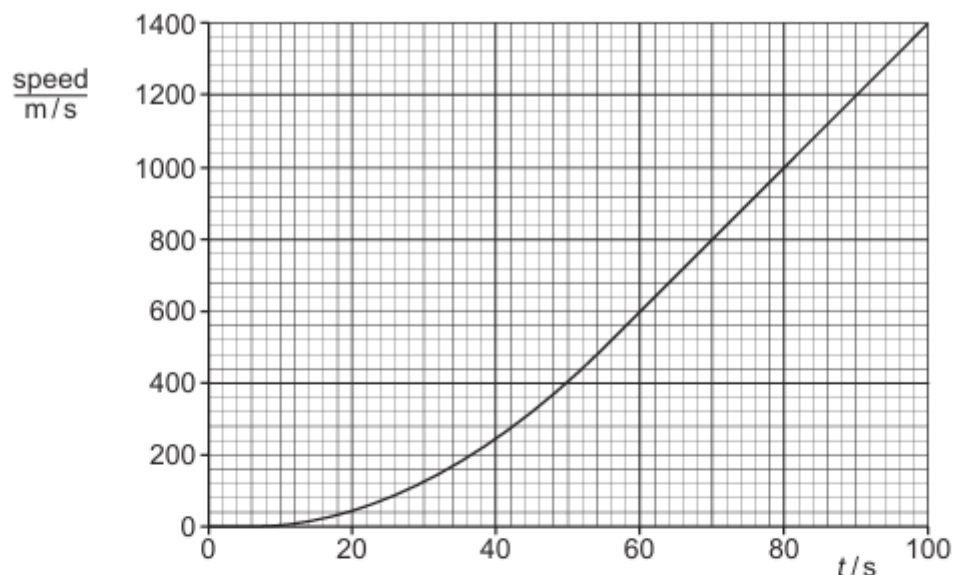


Fig. 9.1

- (a) State the principal energy changes taking place as the rocket accelerates upwards.

.....

[4]

- (b) (i) State the size of the acceleration of the rocket at $t = 0$.

.....[1]

- (ii) State what happens to the acceleration of the rocket between $t = 5$ s and $t = 80$ s.

.....
[2]

- (iii) Calculate the acceleration of the rocket at $t = 80$ s.

acceleration =[2]

- (iv) The total mass of the rocket at $t = 80$ s is 1.6×10^6 kg. Calculate the resultant force on the rocket at this time.

resultant force =[2]

- (v) The total weight of the rocket at $t = 80\text{ s}$ is $1.6 \times 10^7\text{ N}$. Calculate the upward force on the rocket at this time, caused by the burning fuel.

upward force =[1]

- (c) As the rocket burns fuel, it ejects hot gas downwards.

- (i) State *Newton's third law of motion*.

.....
.....[1]

- (ii) Explain how Newton's third law of motion applies to the upward force on the rocket and to the force on the hot gas.

.....
.....
.....[2]

26. M/J/11/21/Q2

Fig. 2.1 shows a sky-diver falling vertically downwards at terminal velocity.

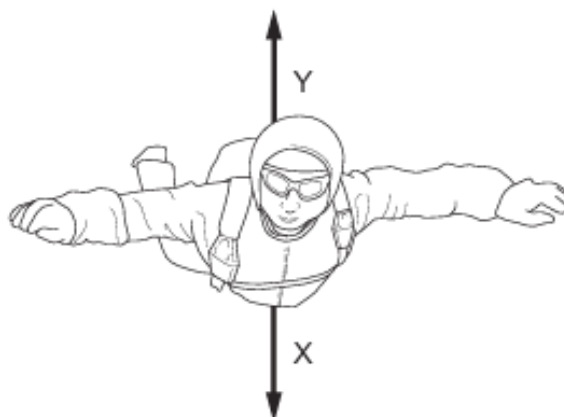


Fig. 2.1

Arrows X and Y show the two main forces acting on the sky-diver.

- (a) (i)** State the name of force X and the name of force Y.

X

Y [1]

- (ii)** Explain why force Y acts upwards.

.....

..... [1]

- (b)** When the sky-diver first started to fall, forces X and Y were unbalanced.

- (i)** Describe and explain the effect of the unbalanced forces on the motion of the sky-diver.

.....

.....

..... [2]

- (ii)** State what happened to the size of force X and the size of force Y as the sky-diver fell and reached terminal velocity.

.....

.....

..... [2]

27. O/N/10/21/Q2

Fig. 2.1 shows the planet Venus orbiting the Sun in a circular orbit at constant speed.

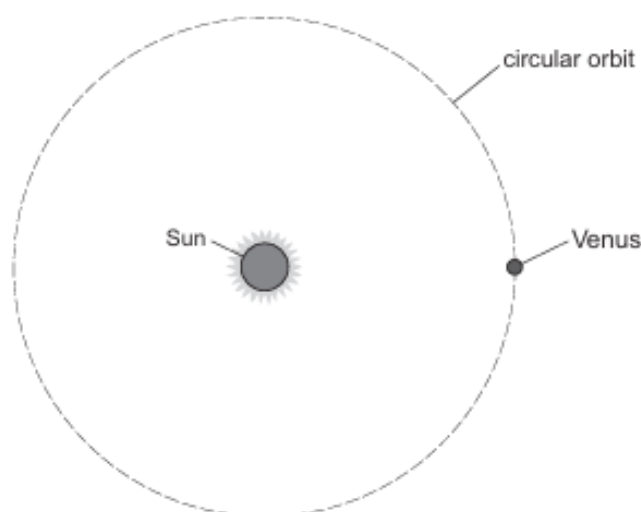


Fig. 2.1 (not to scale)

A force acting on Venus keeps it moving in a circle.

- (a)** State the name of this force and the object that causes it.

.....
 [2]

- (b)** **(i)** On Fig. 2.1, draw an arrow through Venus that represents the direction of its acceleration. [1]
(ii) The size of this acceleration is $9.7 \times 10^{-3} \text{ m/s}^2$ and the mass of Venus is $4.9 \times 10^{24} \text{ kg}$. Calculate the size of the force acting on the planet.

force = [2]

- (c)** As time passes, the planet Venus moves a considerable distance around the circular orbit. State why no work is done by the force acting on the planet as it moves.

.....

 [2]

28. M/J/10/22/Q9

Fig. 9.1 shows a car braking on a road and coming to rest.



Fig. 9.1

(a) Explain what is meant by

(i) the *thinking distance*,

.....
.....[1]

(ii) the *braking distance*.

.....
.....[1]

(b) An engineer conducts a test on the car and finds that the braking distance is greater when the car is fully loaded than when it is unloaded.

(i) Apart from the road conditions, state what must be kept the same in the test.

.....[1]

(ii) Explain why the car has a greater braking distance when fully loaded.

.....
.....[1]

(c) State and explain how one road condition affects the braking distance of the car. Use ideas about friction in your answer.

.....
.....[2]

- (d) Explain how wider tyres affect the pressure of the car on the surface of the road.

.....
 [1]

- (e) The car has a total mass of 900 kg and is travelling at 20 m/s. At time $t = 0$, the driver sees an accident ahead. He applies the brakes at $t = 0.60$ s to stop the car. After the brakes are applied, the car comes to rest in a further 4.0 s.

- (i) Calculate the deceleration of the car as it brakes.

deceleration = [2]

- (ii) Calculate the braking force acting on the car.

force = [2]

- (iii) On Fig. 9.2, draw a speed-time graph for the car as it brakes.

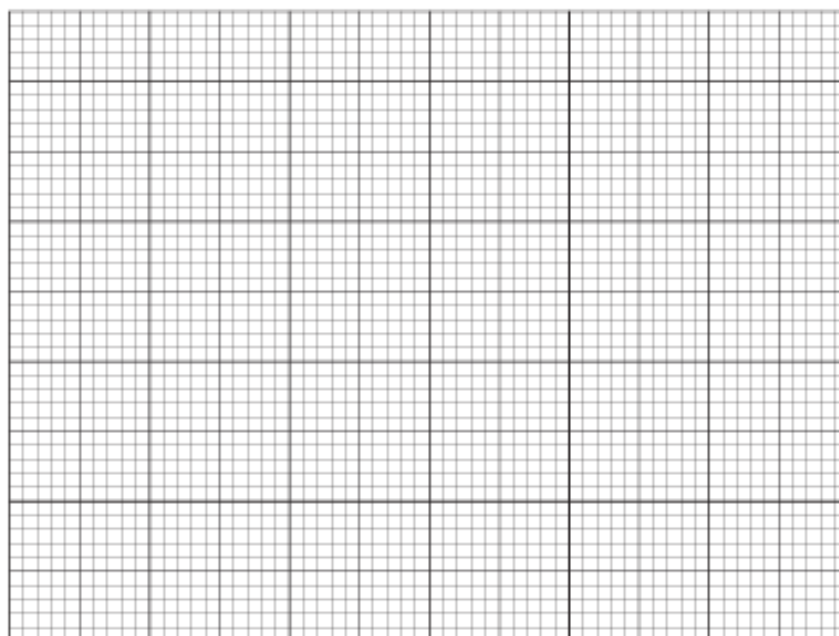


Fig. 9.2

[3]

- (iv) State how your graph in (iii) can be used to find the total distance travelled by the car.

.....
 [1]

ANSWERS

1.

Question	Answer	Marks
2(a)	change / increase of velocity (with time)	B1
	change / increase of velocity per unit time or rate of change / increase of velocity	B1
2(b)(i)	force $\times$ distance moved <u>in the direction of the force / forwards</u>	B1
2(b)(ii)	(work done $\Rightarrow$) $5.4 \times 10^4 \times 280$	C1
	1.5×10^7 (J)	A1

Question	Answer	Marks
2(b)(iii)	$E_k = \frac{1}{2}mv^2$ or ($m \Rightarrow$) $2 \times E_k / v^2$	C1
	($m \Rightarrow$) $2 \times 3.2 \times 10^6 / 12^2$ or $2 \times 3.2 \times 10^6 / 144$	C1
	4.4×10^4 (kg)	A1

2.

Question	Answer	Marks
1(a)	speed is constant for 10 s / at first	B1
	deceleration - has (average) value 0.39 m/s^2 - lasts for 40 s - is non-uniform	B1
1(b)(i)	friction (with road) and <u>air</u> resistance or (air) drag	B1
1(b)(ii)	resistive force(s) change / decrease (as speed decreases) or (graph shows) deceleration is not constant / decreases or non-uniform decrease in speed	B1
1(c)(i)	(momentum $\Rightarrow$) mv in any form algebraic or numerical	C1
	8400 and kg m / s or Ns	A1
1(c)(ii)	($F \Rightarrow$) $\Delta mv / t$ or $F = ma$ in any form algebraic or numerical	C1
	840 (N)	A1

3.

Question	Answer	Marks
7(a)	acceleration / increase in speed and gravity / weight mentioned	B1
	air resistance / friction increases (with speed)	B1
	terminal / constant velocity reached or air resistance (becomes) equal to accelerating force	B1
7(b)	(distance travelled =) area under the graph or $(x =) \frac{1}{2}(u + v)t$ or $\frac{1}{2} \times 15.2 \times 4.0$	C1
	$35 - (\frac{1}{2} \times 15.2 \times 4.0)$ or 30.4 or 30	C1
	4.6 m	A1
7(c)(i)	(magnitude of deceleration =) gradient or Dv / Dt or $15.2 / 4.0$ or equivalent numbers (from graph) in same ratio	C1
	$15.2 / 4.0$ or equivalent numbers in same ratio	C1
	3.8 m/s^2	A1
7(c)(ii)	$(F =) ma$ or $(46 + 9.0) \times 3.8$ or 55×3.8	C1
	210 N	A1
7(c)(iii)	<u>from</u> kinetic energy (and from gravitational potential energy)	B1
	<u>to</u> thermal energy and no intermediate energy	B1
7(d)	due to his mass or mass mentioned	B1
	(his) mass resists a change to its state of motion	B1

4.

Question	Answer	Marks
1(a)	(distance =) area (under graph) or 77×56 or $\frac{1}{2} \times 77 \times 70$ or 4300 or 2700 or $\frac{1}{2} \times 77 \times (56 + 126)$	C1
	$77 \times 56 + (\frac{1}{2} \times 77 \times 70)$ or $4300 + 2700$ or $\frac{1}{2} \times 77 \times (56 + 126)$	C1
	7000 m or $7.0 \times 10^3 \text{ m}$ or 7.0 km	A1
1(b)(i)	$(a =) \Delta v / \Delta t$ or $77 / 70$ (etc.)	C1
	1.1 m/s^2	A1
1(b)(ii)	$(F =) ma$ or $3.8 \times 10^5 \times 1.1$	C1
	$4.2 \times 10^5 \text{ N}$	A1

5.

Question	Answer	Marks
7(a)(ii)	area under curve or formula for area of a triangle / trapezium seen or working shown used	B1
	(accelerating) 80 (m) or (at constant speed) 160 (m) or (decelerating) 32 (m) seen	C1
	(total distance) 270 or 272 m	A1
7(a)(iii)	(average speed) = <u>total</u> distance / (total) time algebraic or numerical	C1
	11 m / s	A1
7(a)(iv)	horizontal line drawn from 0 to 24 s at any level or horizontal line at the value calculated in 7(a)(iii) for at least 16 s	B1
	horizontal line from 0 to 24 s at the value calculated in 7(a)(iii)	B1
7(b)(i)	27	B1
7(b)(ii)	(statement that thinking) distance is proportional to speed or thinking distance increases uniformly with speed or time = (thinking) distance / speed and stated to be equal / not affected or ratio of (thinking) distance to speed is constant or (thinking) distance and speed both increase by same ratio or same change in (thinking) distance for equal changes in speed	C1
	at least two values thinking distance + speed (or inverse) calculated or all values of thinking distance/speed stated to be 0.45 (or inverse 2.2) or proportionality shown by at least two values or double the speed doubles the (thinking) distance or <u>each 20 (km / h) / same</u> increase in speed covers an extra 9 (m) distance	A1
7(b)(iii)	<i>thinking distances</i> increases	B1
	<i>braking distances</i> no change	B1

6.

7(c)	air resistance and weight / gravitational force mentioned	B1
	as speed increases air resistance increases	B1
	<u>resultant</u> / <u>net</u> force decreases or acceleration decreases	B1
	(eventually) air resistance = weight (at terminal velocity) / resultant force is zero	B1
	(at terminal velocity / eventually) zero acceleration / constant speed	B1

7.

Question	Answer	Marks
2(a)	800 N c.a.o.	B1
2(b)(i)	($F =$) ma or 80×0.50	C1
	40 N	A1
2(b)(ii)	840 N	B1
2(c)	increasing gradient from origin to $t = 4.0$ s	B1
	decreasing gradient from end of first curve	B1
	line horizontal at origin or at $t = 8.0$ s (and beyond)	B1

8.

Question	Answer	Marks
2(a)	(a vector quantity has) magnitude and direction	B1
2(b)(i)	the wall exerts a force on swimmer	B1
	opposite in direction (to force of swimmer on wall) or to the right	B1
	equal in size (to force of swimmer on wall) or (resultant) force causes acceleration or force on object Y due to X is equal and opposite to force on object X due to Y	B1

Question	Answer	Marks
2(b)(ii)	(water) resistance / resistive force / friction (on swimmer)	B1
	(water) resistance / backward force / resistive force / friction / opposing force increases (with speed / time)	B1
	(at constant speed) forward force is equal to resistive / backward force or no resultant force or forces balance	B1

9.

Question	Answer	Marks
8(a)(i)	weight of horse in mid-section of horse and labelled	B1
	force upwards on at least two legs and labelled	B1
	force downwards on saddle area labelled <i>F</i>	B1
8(a)(ii)	forces equal and opposite	B1
	acts <u>on rider</u> (upwards from horse)	B1

10.

Question	Answer	Marks
1(a)(i)	accelerating or increasing speed	B1
	decreasing acceleration or speed increasing at a decreasing rate or reaches a constant speed	B1
1(a)(ii)	<u>initially</u> acceleration is due to force of gravity / weight (only) or force of gravity / weight larger than air resistance from ($t = 0$ to ≈ 7 s)	B1
	air resistance increases (with speed)	B1
	resultant force decreases / becomes zero or forces balance or air resistance equals weight	B1
1(b)	two different pairs of co-ordinates from <u>straight-line</u> section of graph seen	C1
	correct use of co-ordinates in a division	C1
	48 – 52 m / s	A1

11.

Question	Answer	Marks
2(a)	it / a scalar does not have a <u>direction</u>	B1
2(b)	<u>displacement</u> and <u>force</u> underlined	B1
2(c)(i)	direction (of motion) is changing	B1
	velocity is changing (with time) or it is accelerating (which requires a resultant force)	B1
2(c)(ii)	gravitational attraction / field of the earth	B1

12.

Question	Answer	Marks
7(a)	(between $t = 0$ and $t = 10$ s) it is accelerating or speed / velocity increasing	B1
	between $t = 35$ s and $t = 40$ s, it is decelerating or speed / velocity decreasing	B1
	magnitude of acceleration less than magnitude of deceleration or between $t = 0$ and $t = 10$ s starts from rest / zero speed or between $t = 35$ ms and $t = 40$ s it finishes at rest / with zero speed	B1
7(b)(i)	two pairs of co-ordinates from the straight-line section	C1
	appropriate division using candidate's coordinates	C1
	$11.7 \leq \text{speed} \leq 12.0$ m / s	A1
7(b)(ii)	<u>total</u> distance / <u>total</u> time or (distance =) 390 (m) and 39 (s) $\leq$ (time =) $\leq$ 40 (s)	C1
	9.7 to 10.0 m / s	A1
7(c)(i)	it / velocity has a direction or is a vector or depends on displacement (rather than distance)	B1
7(c)(ii)	answers need not be in this order 1 0 and 10 (s) and forwards / in the direction of movement	B1
	2 21 (s) and 24 (s)	B1
	perpendicular to the motion / to the direction of movement	B1
	3 35 (s) and 40 (s) and backwards / opposite to the direction of movement	B1
7(d)(i)	speed (of bus) varies / changes / decreases / increases	B1
7(d)(ii)	any time on Fig. 1.1 from $10 \text{ s} \leq t \leq 35 \text{ s}$ labelled M	B1

13.

Question	Answer	Marks
2(a)	96 m	B1
2(b)(i)	increases	B1
	time unchanged but speed faster	B1
2(b)(ii)	decreases	B1
	(force of) <u>friction</u> larger	B1

14.

Question	Answer	Marks
1(a)	78 m	B1
1(b)	(starting to) apply the brakes	B1
1(c)(i)	time taken to think / react is the same (and speed is the same)	B1
1(c)(ii)	friction is lower or less backwards / resistant force or car slides or skids / tyres do not grip road / car aquaplanes	B1
	takes longer time to stop or acceleration is less	B1

15.

9(b)(i)	distance travelled before brakes applied (after seeing emergency)	B1
9(b)(ii)	any 2 of - poor tyre surface - poor road conditions, e.g. rain, ice - high speed (of car) or good aerodynamic car shape or tail wind - large mass / weight of car - driver applies less force to pedal / less braking force	B1B1

16.

Question	Answer	Marks
9(a)(i)	amount / measure / quantity of matter / substance (in a body) or (property of a body) that resists (change in) motion	B1
9(a)(ii)	force / pull of <u>gravity</u> or in a gravitational field	B1
9(b)	(a =) F / m algebraic or numerical or 400 (N) seen	C1
	400 / 60	C1
	6.7 m/s ²	A1
9(c)(i)	initial curve of correct shape from origin at 0 to 30 s	B1
	horizontal line at 44 m/s from 30 s to 40 s	B1
	speed drops from 44 to 5 m/s in a time of 4 s	B1
	constant speed of 5 m/s until 200 s (and then to zero)	B1
9(c)(ii)	(a =) $(v - u) / t$ algebraic or numerical	C1
	9.7 or 9.8 m/s ²	A1
9(d)(i)	upwards force / air resistance = weight / downward force	B1
9(d)(ii)	increase in air resistance / upwards force (from parachute)	B1
	upwards / force / air resistance larger than weight / downwards force or resultant force upwards	B1
9(d)(iii)	air resistance decreases (as speed reduces) until it equals weight or (with parachute open) air resistance = weight at lower speed	B1

17.

(a)	$(F =) ma$ or 35×2.6	C1
	91 N	A1
(b) (i)	straight line from origin to dashed line / t_1	B1
	gradually decreasing gradient until the line is horizontal	B1
(b) (ii)	area <u>under</u> the line or area between line and x-axis	B1

18.		
(a)	(i) equal and opposite forces or no resultant force	B1
	(ii) moves / slides	B1
	constant velocity	B1
	or constant speed and direction	
(b)	(resultant force) 15 (N) seen	C1
(b)	(a=) F / m in any form numerical or algebraic	C1
	$0.60 \text{ m} / \text{s}^2$	A1
19.		
(a)	A – contact or reaction force B – driving force C – force of gravity D – air resistance and friction	B1
(b)	(i) 800 kg	B1
	(ii) 400 N	B
	(iii) (a =) F / m formula or with numbers	C1
	$0.50 \text{ m} / \text{s}^2$	A1
(c)	change in $v = at$ in words or numbers	C1
	$12 \text{ m} / \text{s}$	A1
20.		
(a)	kinetic energy at start	B1
	thermal energy/heat energy/ internal energy at end	B1
(b)	(i) $0.4(0) \text{ s}$	B1
	(ii) (d=) $s \times t$ numerical or algebraic	C1
	2.8 m	A1
	(iii) area under graph (between 0.4 and 2.4 s) or time (difference) $\times$ <u>average</u> speed or $\frac{1}{2} \times$ time (difference) $\times$ initial speed	B1
(iv)	horizontal line from (0,5) to (0.4,5)	B1
	line showing braking with same gradient as original line	B1

- (v) less friction B1
 less deceleration **or** graph less steep B1
or less force backwards/less force opposing motion
or same KE lost/work done by friction
 longer time to stop B1
or larger area under (speed-time) graph
or work = force x distance applied correctly

21.

- (a) (i) **D and either** lorry accelerates (forward) **or** resultant force is forward [B1]
 (ii) air resistance **or** (air) drag **or** friction (between tyres and road) [B1]
 (b) (i) 30 000 kg [B1]
 (ii) (a=) F/m algebraic in any form or numerical [C1]
 $0.5(0) \text{ m/s}^2$ [A1]
 (c) direction or velocity is changing **or** acceleration or force is sideways or towards centre (of circle) [B1]

22.

- (a) (i) 850 N B1
 (ii) $KE = PE/mgh$ **or** $mgh = 5.5 \times 10^4$ C1
 $65/64.7(0588) \text{ m}$ A1
 (b) $WD = Fx$ **or** KE/x **or** $5.5 \times 10^4/33$ **or** $v = 35(.97)$ **and** $a = 19(.60)$ **and** $F = ma$ C1
 $1700/1670/1667/1666.7 \text{ N}$ A1

23.

- (a) (i) speed and mass B2
 (ii) 1 speed and direction B1
or distance/time and direction
or displacement/time
 2 direction changes B1
 (iii) force of gravity from/towards Earth B1
 force is centripetal B1
or at right angles to motion/velocity
 (b) (i) 450 000 N B1
 (ii) (a =) F/m or 50 000/40 000 C1
 1.25 m/s^2 A1

- 24.
- (a) a force B1
when objects slide over/rub one another
OR opposes (relative) **motion/movement** B1
- (b) (i) constant/uniform speed OR constant/uniform velocity OR zero acceleration B1
- (ii) ($F =$) ma seen in any form numerical or algebraic C1
1200 (N) OR 6200 N seen C1
3800 N A1
- (iii) Force B increases OR backwards force/resistance/friction/drag increases M1
as speed/velocity increases A1
- 25.
- (a) **at start:** chemical (potential) energy B1
at end: PE/GPE/gravitational energy B1
at end: heat/thermal/internal energy B1
at end: KE **or** intermediate KE from chemical energy B1
- (b) (i) 0 B1
- (ii) it increases B1
to constant value B1
- (iii) gradient **or** $(v-u)/t$ **or** $(1400-600)/40$ **or** other correct numbers C1
 20 m/s^2 A1
- (iv) ($F =$) ma **or** $1.6 \times 10^6 \times 20$ C1
 $3.2 \times 10^7 \text{ N}$ A1
- (v) $4.8 \times 10^7 \text{ N}$ B1
- (c) (i) to every action there is an equal and opposite reaction
or forces act in pairs of equal size **and** in opposite directions/on
different bodies B1
- (ii) downward force on gas B1
equal and opposite to upward/(b)(v) force (on rocket) B1
- 26.
- (a) (i) X weight // (force of) gravity // gravitational (force)
and Y air resistance // (air) drag // wind resistance // **air** friction B1
- (ii) (Y) opposes motion // diver moves down // air molecules hit faster from below B1
- (b) (i) accelerates // falls faster // speed/velocity greater B1
X is larger than Y // resultant/net force downwards B1
- (ii) **two** of:
X stays the same // Y increases // forces become equal (and opposite) B2

- 27.
- (a) gravitational/centripetal (pull/attraction) of the **Sun** B1
B1
- (b) (i) arrow touching Venus towards centre/left (must pass through Sun if extended) B1
- (ii) ($F =$) ma or $4.9 \times 10^{24} \times 9.7 \times 10^{-3}$ C1
 $4.8(4.753) \times 10^{22}$ N A1
- (c) direction of movement $\perp$ direction of force C1
force/it is perpendicular/at right angles to distance moved **or** does not move **any** distance in direction of force A1
- 28.
- (a) (i) distance travelled while thinking/in reaction time or before braking starts B1
- (ii) distance travelled while brakes applied/car decelerates B1
- (b) (i) speed (of cars) or same force/pressure on pedal or same braking force or same tyres or condition of brakes B1
- (ii) greater inertia/kinetic energy/momentum or smaller deceleration/acceleration B1
- (c) any road condition, e.g. icy, wet, poor surface, slippery/smooth/rough surface and its correct effect on distance B1
correct explanation that refers to friction e.g. more friction when dry B1
- (d) pressure low(er) (with larger area) B1
- (e) (i) $a = v/t$ any algebraic or numerical value e.g. 20/4; 20/3.4; 20/4.6; 20/0.6 C1
 $5(.0)\text{m/s}^2$ A1
- (ii) $F = ma$ algebraic or numerical e.g. $900 \times$ (i) C1
4500 N e.c.f. (i) A1
- (iii) correct axes labelled with quantity and/or unit B1
horizontal line at 20 m/s from 0 to 0.6 s B1
straight line from end of horizontal section or from (0.6,20) to (4.6,0) or (4,0) B1
- (iv) area (under graph or of trapezium) B1